Detailed derivation of the quantum theory of inverse bremsstrahlung heating of the free electrons of a plasma

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These notes derive, with every intermediate result written out, the rate at which a laser field heats the free electrons of a fully ionized plasma by inverse bremsstrahlung. The starting point is the Schrödinger equation of an electron in the field of the laser and of the ions. The electron in the laser field is described exactly by the Volkov states [2], the ion potential is treated in Born approximation as in Bunkin and Fedorov [3], and the occupation numbers obey a kinetic equation that is derived with the operator method of the quantum Boltzmann equation paper [4]. The classical treatment of Atzeni and Meyer-ter-Vehn [1] is the baseline. Every boxed result is checked numerically by `detailed_derivation_check.py`*, which Appendix C describes.*

For notation, Gaussian units are used throughout, as in [1]. The electron has charge $-e$ and mass m. The ions have charge Ze, are infinitely heavy and sit at fixed positions $\mathbf{R}_j$ with density n_i, and the plasma is neutral, $n_e = Zn_i$. The temperature T is in kelvin and k_B is written explicitly. The laser field is uniform and linearly polarized along the unit vector $\hat{\mathbf{e}}$. The electrons are in a box of volume $\mathcal{V}$ with periodic boundary conditions, a momentum transfer is written $\hbar\mathbf{k}$, and the integer ℓ counts the photons absorbed in one collision, with $\ell < 0$ for emission. The mean occupation of the electron state of momentum $\mathbf{p}$, per spin state, is $f_{\mathbf{p}}$. It is the $n_{\mathbf{p}}$ of the quantum Boltzmann equation paper. The Fourier transform of a function of position is $\tilde{v}(\mathbf{k}) = \int v(\mathbf{r})\, e^{-i\mathbf{k}\cdot\mathbf{r}}\, \mathrm{d}^3 r$. A term that is struck through vanishes, for the reason given at that step. A boxed equation is a result that is used later. An equation of another document is cited by its own number followed by the reference, as in Eq. (11.12) [1], and an equation number without a reference is one of these notes.

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1 Why a quantum theory of inverse bremsstrahlung?

(1) Inverse bremsstrahlung is the absorption of laser photons by free electrons while they collide with ions. In laser-driven inertial fusion it is the preferred absorption mechanism, because the absorbed energy goes into the thermal electrons and stays where it was absorbed [1, Sec. 11.2].
(2) A free electron alone cannot absorb a photon. Let an electron of momentum $\mathbf{p}$ absorb a photon of energy $\hbar\omega$ and momentum $\hbar\mathbf{k}_L$. Conservation of energy and momentum requires

$$\frac{|\mathbf{p}+\hbar\mathbf{k}_L|^2}{2m} - \frac{p^2}{2m} = \hbar\omega \quad \Longrightarrow \quad \frac{\omega}{k_L} = \frac{\mathbf{p}\cdot\hat{\mathbf{k}}_L}{m} + \frac{\hbar k_L}{2m}. \tag{1.1}$$

The left side is the phase velocity of the transverse wave. In a plasma it is c/n_r with the refractive index $n_r = (1-\omega_p^2/\omega^2)^{1/2} < 1$, so it exceeds c. The right side is the velocity of a nonrelativistic electron plus $\hbar k_L/2m = \hbar\omega n_r/(2mc)$, which is of order $10^{-5}c$ for optical photons. The two sides cannot be equal, and a third body has to take up the momentum. In a fully ionized plasma that body is an ion.
(3) In the classical theory [1, Sec. 11.1.1] the electrons oscillate in the laser field and a friction force $-\nu_e m\mathbf{v}$ represents their collisions with the ions. The absorption coefficient follows from the imaginary part of the dielectric function, and ν_e is an input. Its Coulomb logarithm needs two cutoffs. For $k_BT_e \geq 10$ eV the lower one is the de Broglie wavelength of the electron [1, Eq. (10.136)], which is a quantum statement put in by hand.
(4) The photon energy enters the classical theory twice, through the factor $1-e^{-\hbar\omega/k_BT}$ for stimulated emission [1, Eq. (11.34)] and through the Gaunt factor of the free–free cross section [1,

Sec. 10.6]. Table 1 shows that $u = \hbar\omega/k_BT$ ranges from 0.05 at 100 eV to 0.001 at 3 keV for ultraviolet lasers. It is small, but it sets the Coulomb logarithm, as Sec. 8 shows.
(5) At the intensities of inertial fusion the quiver velocity $v_0 = eE_0/m\omega$ is a sizable fraction of the thermal velocity $v_t = (k_BT/m)^{1/2}$, and at 100 eV it is close to it, as Table 1 shows. The absorption is then not linear in the intensity, and several photons can be exchanged in one collision.
(6) A quantum theory removes all of these inputs. The electron in the laser field is described exactly by the Volkov states, the ion potential is treated in Born approximation, and the result holds for any field strength, any photon energy and any degeneracy of the electrons. The Born approximation needs $\eta = Ze^2/\hbar v \ll 1$ [3, Eq. (5)], which Table 1 shows to hold for hydrogen above about 100 eV and for carbon–hydrogen plastic above about 1 keV.

Table 1: Scales of inverse bremsstrahlung for the third harmonic of the Nd:glass laser and for the KrF laser, at the density $n_e = n_c/4$ and the vacuum intensity 10^{15} W cm^{-2}. Here $u = \hbar\omega/k_BT$, η/Z is the Born parameter at the thermal velocity, $\Lambda(u)$ is the unscreened quantum Coulomb logarithm of Eq. (8.4), $\Lambda_{\rm RPA}$ the dynamically screened one of Eq. (10.13), and $\ln\Lambda_e$ is Eq. (10.136) [1].

λ_L (μm)	k_BT (keV)	u	η/Z	v_0/v_t	$\Lambda(u)$	$\Lambda_{\rm RPA}$	$\ln\Lambda_e$
0.351	0.1	0.0353	0.52	0.68	4.15	4.05	4.39
0.351	1	0.0035	0.16	0.21	6.45	6.36	6.69
0.351	3	0.0012	0.10	0.12	7.55	7.45	7.79
0.248	0.1	0.0500	0.52	0.48	3.81	3.71	4.04
0.248	1	0.0050	0.16	0.15	6.11	6.01	6.34
0.248	3	0.0017	0.10	0.09	7.21	7.11	7.44

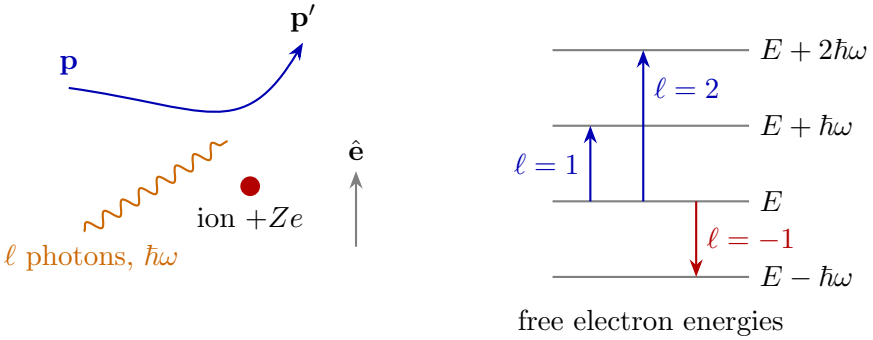


Figure 1: Inverse bremsstrahlung. Left, an electron of momentum $\mathbf{p}$ scatters on an ion into $\mathbf{p}'$ while absorbing ℓ photons from the laser, which is polarized along $\hat{\mathbf{e}}$. The ion takes up the momentum $\mathbf{p} - \mathbf{p}'$. Right, the energy of the free electron changes by $\ell\hbar\omega$, with $\ell < 0$ for stimulated emission.

The plan of these notes. Section 2 sets out the classical theory that the quantum theory must reproduce. Section 3 writes the Hamiltonian and the quiver motion. Section 4 derives the Volkov states. Section 5 derives the multiphoton transition rates in Born approximation and the cross section of Bunkin and Fedorov. Section 6 derives the kinetic equation for the occupation numbers with the operator method of the quantum Boltzmann equation paper, and from it the heating rate. Section 7 writes the heating rate through the Lindhard function. Section 8 evaluates it in closed form for weak fields, and Section 9 connects it to the bremsstrahlung cross section, the Gaunt factor and Kirchhoff's law. Section 10 derives the same rate for the interacting electron gas and adds

dynamic screening. Section 11 treats strong fields, and Section 12 derives the kinetic equation of an isotropic distribution and its Fokker–Planck limit. Section 13 collects the results. Appendix A reduces the results to the classical treatment of Atzeni and Meyer-ter-Vehn equation by equation, Appendix B proves the mathematical identities, and Appendix C lists the numerical checks.

2 The classical baseline

2.1 The Drude model

This subsection follows [1, Sec. 11.1.1] and uses its convention, in which a perturbation varies as $e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}$ and the physical quantity is the real part.
(1) The electron velocity obeys Newton's equation with a friction force,

$$-i\omega m\mathbf{v} = -e\mathbf{E} - \nu m\mathbf{v}, \tag{2.1}$$

which is Eq. (11.7) [1]. Collect the terms in $\mathbf{v}$,

$$m(\nu - i\omega)\,\mathbf{v} = -e\mathbf{E} \quad \longrightarrow \quad \mathbf{v} = -\frac{e\mathbf{E}}{m(\nu - i\omega)} = -\frac{ie\mathbf{E}}{m(\omega + i\nu)}, \tag{2.2}$$

where the last step uses $\nu - i\omega = -i(\omega + i\nu)$ and $1/i = -i$.
(2) The current density and the conductivity are

$$\mathbf{j} = -en_e\mathbf{v} = \frac{ie^2 n_e}{m(\omega + i\nu)}\,\mathbf{E}, \qquad \sigma = \frac{ie^2 n_e}{m(\omega + i\nu)} = \frac{e^2 n_e}{m(\nu - i\omega)}. \tag{2.3}$$

Eq. (11.8) [1] prints $\mathbf{j}_1 = e^2 n_0 \mathbf{E}_1/[im_e(\omega + i\nu_e)]$, which is the negative of Eq. (2.3). The sign of Eq. (2.3) is the one that Eq. (11.9) [1] requires, as the next step shows.
(3) With the dielectric tensor $\varepsilon = 1 + (4\pi i/\omega)\sigma$ of Eq. (11.4) [1],

$$\boxed{\varepsilon_T(\omega) = 1 + \frac{4\pi i}{\omega}\,\frac{ie^2 n_e}{m(\omega + i\nu)} = 1 - \frac{\omega_p^2}{\omega(\omega + i\nu)}, \qquad \omega_p^2 = \frac{4\pi e^2 n_e}{m},} \tag{2.4}$$

which is Eq. (11.9) [1]. Its imaginary part is

$$\operatorname{Im}\varepsilon_T = \frac{\omega_p^2}{\omega}\,\operatorname{Im}\frac{-1}{\omega + i\nu} = \frac{\omega_p^2}{\omega}\,\frac{\nu}{\omega^2 + \nu^2} = \frac{\omega_p^2 \nu}{\omega^3}\Big[1 + O\Big(\frac{\nu^2}{\omega^2}\Big)\Big]. \tag{2.5}$$

(4) The power absorbed per unit volume is the cycle average of $\mathbf{j}\cdot\mathbf{E}$. For the field $\operatorname{Re}[\mathbf{E}_1 e^{-i\omega t}]$ with the real amplitude $E_0 = |\mathbf{E}_1|$,

$$\langle \mathbf{j}\cdot\mathbf{E}\rangle_t = \tfrac{1}{2}\operatorname{Re}\sigma\, E_0^2 = \frac{\omega}{8\pi}\operatorname{Im}\varepsilon_T\, E_0^2 = \frac{e^2 n_e}{2m}\,\frac{\nu}{\nu^2 + \omega^2}\,E_0^2. \tag{2.6}$$

With the quiver velocity $v_0 = eE_0/m\omega$ that Section 3.1 derives,

$$\boxed{P = \frac{\omega}{8\pi}\operatorname{Im}\varepsilon_T\, E_0^2 = n_e\,\nu\,\frac{mv_0^2}{2}\Big[1 + O\Big(\frac{\nu^2}{\omega^2}\Big)\Big].} \tag{2.7}$$

The power is the collision rate times the kinetic energy $mv_0^2/2$ of the quiver amplitude. Every quantum result of these notes is written in this form, which defines the effective collision frequency $\nu_{\text{eff}} \equiv P/(n_e m v_0^2/2)$.

(5) The wave number of a transverse wave obeys $k^2 = (\omega^2/c^2)\varepsilon_T$, Eq. (11.5) [1]. To first order in $\operatorname{Im}\varepsilon_T$,

$$k = \frac{\omega}{c}\Big[(\operatorname{Re}\varepsilon_T)^{1/2} + \frac{i\operatorname{Im}\varepsilon_T}{2(\operatorname{Re}\varepsilon_T)^{1/2}}\Big], \tag{2.8}$$

and the intensity, which varies as $|E|^2 \propto e^{-2\operatorname{Im}k\,x}$, decays with the coefficient

$$\boxed{K = 2\operatorname{Im}k = \frac{\omega}{c}\frac{\operatorname{Im}\varepsilon_T}{(\operatorname{Re}\varepsilon_T)^{1/2}} = \frac{\nu_{\text{eff}}}{c}\frac{\omega_p^2}{\omega^2}\frac{1}{n_r}, \qquad n_r = \Big(1 - \frac{\omega_p^2}{\omega^2}\Big)^{1/2},} \tag{2.9}$$

which is Eq. (11.12) [1]. The same coefficient follows as the ratio of P to the energy flux $I = cn_rE_0^2/8\pi$ of the wave in the plasma.

2.2 The collision frequency and the Coulomb logarithm

(1) The electron collision time of Eq. (10.133) [1] is

$$\frac{1}{\tau_e} = \frac{4\sqrt{2\pi}}{3}\frac{n_iZ^2e^4}{m^{1/2}(k_BT)^{3/2}}\ln\Lambda_e \equiv \nu_T\ln\Lambda_e, \qquad \boxed{\nu_T = \frac{4\sqrt{2\pi}}{3}\frac{n_iZ^2e^4}{m^{1/2}(k_BT)^{3/2}}.} \tag{2.10}$$

These notes write every collision frequency as ν_T times a dimensionless logarithm Λ, so that Λ can be compared with $\ln\Lambda_e$ directly.
(2) The Coulomb logarithm of Eq. (10.136) [1] is $\ln\Lambda_e = 7.1 - 0.5\ln n_e + \ln T_e$ with n_e in 10^{21} cm^{-3} and T_e in keV. It is the logarithm of the ratio of the Debye length to the de Broglie wavelength $\hbar/2mv$ at the velocity $mv^2/2 = 3k_BT/2$. The lower cutoff is quantum mechanical, and the theory does not derive it.
(3) Two further points are not settled by the classical theory. The collision time τ_e belongs to the slow relaxation of the electrons, while the absorption of light at $\omega \gg \nu$ involves the collisions within one period of the field. Dawson and Oberman [6] and Johnston and Dawson [7] showed that the upper cutoff is then v_t/ω rather than the Debye length. The quantum theory produces both cutoffs, Eq. (8.5).

2.3 The free–free cross section

(1) Kramers' cross section for the emission of a photon of frequency ν by an electron of velocity v in the field of an ion is [8]

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\nu} = \frac{32\pi^2}{3\sqrt{3}}\frac{Z^2e^6}{m^2c^3v^2h\nu}, \tag{2.11}$$

Eq. (10.89) [1]. Quantum corrections are written as a Gaunt factor g that multiplies it.
(2) Averaged over a Maxwellian, Eq. (2.11) gives the emission coefficient of Eq. (10.91) [1], and Kirchhoff's law gives the free–free opacity of Eq. (10.92) [1]. Corrected for stimulated emission and refraction, the absorption coefficient of the laser light is Eq. (11.34) [1],

$$K = \rho\,\kappa_\nu^{\text{ff}}\big(1 - e^{-h\nu/k_BT_e}\big)\Big/\Big(1 - \frac{\omega_p^2}{\omega_L^2}\Big)^{1/2}. \tag{2.12}$$

Atzeni and Meyer-ter-Vehn state that this agrees with Eq. (2.9) in its dependence on density, temperature and wavelength, and in its numerical factor within a few percent. The dependences agree, but with $g = 1$ in Eq. (10.92) [1] and $\ln\Lambda_e = 8$ the numerical factors differ by a factor

4.4. Section 9 derives the Gaunt factor that removes the difference, and Appendix A.5 gives the numbers.

In Summary: The classical theory takes three inputs. The collision frequency enters the Drude model as a parameter, its Coulomb logarithm needs a lower cutoff that is quantum mechanical, and the free–free route needs a Gaunt factor. The quantum theory of the following sections supplies all three, and its result is written as $P = n_e \nu_{\rm eff}\, m v_0^2/2$ with $\nu_{\rm eff} = \nu_T \Lambda$.

3 The Hamiltonian

3.1 The laser field and the quiver motion

(1) The laser field is uniform over the region of a collision. Its electric field and a vector potential that produces it, with zero scalar potential, are

$$\mathbf{E}(t) = E_0\, \hat{\mathbf{e}} \cos\omega t, \qquad \mathbf{A}(t) = -\frac{cE_0}{\omega}\, \hat{\mathbf{e}} \sin\omega t, \tag{3.1}$$

since $-c^{-1}\partial_t \mathbf{A} = E_0 \hat{\mathbf{e}} \cos\omega t$. The validity of the uniform field is the subject of Section 3.4.
(2) A classical electron obeys $m\dot{\mathbf{v}} = -eE_0 \hat{\mathbf{e}} \cos\omega t$. Integrating once and twice,

$$\mathbf{v}(t) = \mathbf{v}_d - v_0\, \hat{\mathbf{e}} \sin\omega t, \qquad \mathbf{r}(t) = \mathbf{r}_d + \mathbf{v}_d t + a_0\, \hat{\mathbf{e}} \cos\omega t, \tag{3.2}$$

where $\mathbf{v}_d$ and $\mathbf{r}_d$ are constants. The oscillating parts define the quiver displacement and the quiver velocity,

$$\boxed{\begin{aligned} &\boldsymbol{\alpha}(t) = \mathbf{a}_0 \cos\omega t, \qquad \mathbf{v}_{\rm osc}(t) = \dot{\boldsymbol{\alpha}}(t) = -v_0\, \hat{\mathbf{e}} \sin\omega t, \qquad \mathbf{a}_0 = a_0 \hat{\mathbf{e}}, \\ &v_0 = \frac{eE_0}{m\omega}, \qquad a_0 = \frac{v_0}{\omega} = \frac{eE_0}{m\omega^2}, \qquad U_{\rm p} = \left\langle \frac{m\mathbf{v}_{\rm osc}^2}{2} \right\rangle_t = \frac{mv_0^2}{4} = \frac{e^2E_0^2}{4m\omega^2}. \end{aligned}} \tag{3.3}$$

Here $\langle \cdots \rangle_t$ is the average over one period, and $U_{\rm p}$ is the ponderomotive energy. The second derivative is $\ddot{\boldsymbol{\alpha}} = -\omega^2 \boldsymbol{\alpha} = -(e/m)\mathbf{E}(t)$, and the quiver velocity is also

$$\dot{\boldsymbol{\alpha}}(t) = \frac{e}{mc}\mathbf{A}(t). \tag{3.4}$$

(3) For a charge $-e$ the canonical momentum is $\mathbf{p} = m\mathbf{v} - (e/c)\mathbf{A}$. With Eq. (3.1), $(e/c)\mathbf{A} = -mv_0 \hat{\mathbf{e}} \sin\omega t$ and $\mathbf{p} = m\mathbf{v} + mv_0 \hat{\mathbf{e}} \sin\omega t = m\mathbf{v}_d$ is constant. The canonical momentum is the drift momentum, and the kinetic momentum $m\mathbf{v} = \mathbf{p} + (e/c)\mathbf{A} = \mathbf{p} + m\mathbf{v}_{\rm osc}$ carries the quiver motion.

3.2 The Hamiltonian of one electron

(1) The Hamiltonian of one electron in the laser field and in the field of the ions is

$$\boxed{H(t) = \frac{1}{2m}\left[\hat{\mathbf{p}} + \frac{e}{c}\mathbf{A}(t)\right]^2 + V(\hat{\mathbf{r}}), \qquad V(\mathbf{r}) = \sum_{j=1}^{N_i} v(\mathbf{r} - \mathbf{R}_j),} \tag{3.5}$$

where $v(\mathbf{r})$ is the potential energy of the electron in the field of one ion and $N_i = n_i \mathcal{V}$.
(2) For a bare Coulomb ion $v(\mathbf{r}) = -Ze^2/r$, and for a statically screened one $v(\mathbf{r}) = -Ze^2 e^{-\kappa r}/r$. The Fourier transform follows from the angular integral $\int e^{-i\mathbf{k}\cdot\mathbf{r}}\, \mathrm{d}\Omega_r = 4\pi \sin(kr)/kr$,

$$\int \frac{e^{-\kappa r}}{r}\, e^{-i\mathbf{k}\cdot\mathbf{r}}\, \mathrm{d}^3 r = \frac{4\pi}{k} \int_0^\infty \sin(kr)\, e^{-\kappa r}\, \mathrm{d}r = \frac{4\pi}{k}\, \frac{k}{k^2 + \kappa^2}, \tag{3.6}$$

so that

$$\boxed{\tilde{v}(\mathbf{k}) = -\frac{4\pi Z e^2}{k^2+\kappa^2} \xrightarrow{\kappa\to 0} -\frac{4\pi Z e^2}{k^2}.} \tag{3.7}$$

These notes use the bare Coulomb form unless stated otherwise. Screening is treated dynamically in Section 10.

(3) With the Fourier series $v(\mathbf{r}) = \mathcal{V}^{-1}\sum_{\mathbf{k}} \tilde{v}(\mathbf{k}) e^{i\mathbf{k}\cdot\mathbf{r}}$ the potential of all the ions is

$$V(\mathbf{r}) = \frac{1}{\mathcal{V}}\sum_{\mathbf{k}} \tilde{v}(\mathbf{k})\,\rho_i(\mathbf{k})\,e^{i\mathbf{k}\cdot\mathbf{r}}, \qquad \rho_i(\mathbf{k}) = \sum_j e^{-i\mathbf{k}\cdot\mathbf{R}_j}, \qquad \tilde{V}(\mathbf{k}) \equiv \int V(\mathbf{r}) e^{-i\mathbf{k}\cdot\mathbf{r}} \mathrm{d}^3 r = \tilde{v}(\mathbf{k})\rho_i(\mathbf{k}). \tag{3.8}$$

3.3 The oscillating frame of Kramers and Henneberger

The quiver motion can be removed from the kinetic energy by going into the frame that oscillates with the electron [12]. This frame gives the clearest picture of inverse bremsstrahlung, and Section 10 uses it for many electrons.

(1) Let $\hat{T}(t) = \exp[-(i/\hbar)\boldsymbol{\alpha}(t)\cdot\hat{\mathbf{p}}]$, which translates a wave function by $\boldsymbol{\alpha}$, $(\hat{T}\phi)(\mathbf{r}) = \phi(\mathbf{r}-\boldsymbol{\alpha})$. Then $\hat{T}^\dagger V(\hat{\mathbf{r}})\hat{T} = V(\hat{\mathbf{r}}+\boldsymbol{\alpha})$, $\hat{T}$ commutes with $\hat{\mathbf{p}}$, and $i\hbar\,\partial_t\hat{T} = (\dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}})\hat{T}$, because $\boldsymbol{\alpha}\cdot\hat{\mathbf{p}}$ and $\dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}}$ commute.

(2) Write the wave function as $\psi = e^{-iS(t)/\hbar}\hat{T}\phi$ with a real function $S(t)$. Insert it into $i\hbar\partial_t\psi = H\psi$,

$$e^{-iS/\hbar}\hat{T}\big[\dot{S} + \dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}} + i\hbar\partial_t\big]\phi = e^{-iS/\hbar}\hat{T}\big[\hat{T}^\dagger H \hat{T}\big]\phi, \tag{3.9}$$

and multiply by $e^{iS/\hbar}\hat{T}^\dagger$,

$$i\hbar\partial_t\phi = \Big[\frac{\hat{p}^2}{2m} + \frac{e}{mc}\mathbf{A}\cdot\hat{\mathbf{p}} + \frac{e^2A^2}{2mc^2} + V(\hat{\mathbf{r}}+\boldsymbol{\alpha}) - \dot{\boldsymbol{\alpha}}\cdot\hat{\mathbf{p}} - \dot{S}\Big]\phi. \tag{3.10}$$

(3) By Eq. (3.4) the second and fifth terms cancel. Choosing $\dot{S} = e^2A^2/2mc^2$ removes the third, which is independent of position. What remains is

$$\boxed{i\hbar\partial_t\phi = \Big[\frac{\hat{p}^2}{2m} + V\big(\hat{\mathbf{r}}+\boldsymbol{\alpha}(t)\big)\Big]\phi.} \tag{3.11}$$

In this frame the electron is free and the ions oscillate. The ion j sits at $\mathbf{R}_j - \boldsymbol{\alpha}(t)$. A collision is elastic in the rest frame of the ion, which is the laboratory, and the energy that the electron gains in the oscillating frame comes from the motion of the ion relative to that frame. Step (3) of Section 5.6 makes this precise.

3.4 Validity of the uniform field and of nonrelativistic motion

The field of Eq. (3.1) neglects the variation of the laser field over the region of a collision and the magnetic force. Bunkin and Fedorov state the conditions as $v/c \ll 1$ and $\omega x_0/c \ll 1$, where x_0 is the larger of the distance v/ω traveled in one period and the quiver amplitude a_0 [3, Eq. (4)]. Since $\omega a_0 = v_0$, both conditions are $v \ll c$ and $v_0 \ll c$, the nonrelativistic motion of the electron. At 10^{15} W cm^{-2} and $\lambda_L = 0.351$ μm, $v_0/c = 0.0095$, and at 1 keV, $v_t/c = 0.044$.

4 The Volkov states

4.1 The solution

(1) Without the ions the Schrödinger equation is

$$i\hbar\,\partial_t\psi = \frac{1}{2m}\Big[-i\hbar\nabla + \frac{e}{c}\mathbf{A}(t)\Big]^2\psi. \tag{4.1}$$

Since $\mathbf{A}$ does not depend on position, the plane wave $e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}$ is an eigenfunction of the operator on the right at every time. Try $\psi = \mathcal{V}^{-1/2}e^{i\mathbf{p}\cdot\mathbf{r}/\hbar}e^{-i\Phi_\mathbf{p}(t)}$, which gives

$$\hbar\dot\Phi_\mathbf{p} = \frac{1}{2m}\Big[\mathbf{p} + \frac{e}{c}\mathbf{A}(t)\Big]^2. \tag{4.2}$$

(2) Expand the square with Eq. (3.1), where $(e/c)\mathbf{A} = -mv_0\hat{\mathbf{e}}\sin\omega t$,

$$\frac{1}{2m}\Big[\mathbf{p}+\frac{e}{c}\mathbf{A}\Big]^2 = \frac{p^2}{2m} - v_0(\mathbf{p}\cdot\hat{\mathbf{e}})\sin\omega t + \frac{mv_0^2}{2}\sin^2\omega t = E_\mathbf{p} - v_0(\mathbf{p}\cdot\hat{\mathbf{e}})\sin\omega t + U_\mathrm{p}\,(1-\cos 2\omega t), \tag{4.3}$$

with $E_\mathbf{p} = p^2/2m$ and $U_\mathrm{p} = mv_0^2/4$. Integrate from 0 to t with $\int_0^t \sin\omega t'\mathrm{d}t' = (1-\cos\omega t)/\omega$ and $\int_0^t \cos 2\omega t'\mathrm{d}t' = \sin(2\omega t)/2\omega$,

$$\hbar\Phi_\mathbf{p}(t) = (E_\mathbf{p} + U_\mathrm{p})\,t - \mathbf{p}\cdot\mathbf{a}_0\,(1-\cos\omega t) - \frac{U_\mathrm{p}}{2\omega}\sin 2\omega t. \tag{4.4}$$

The Volkov state is therefore [2]

$$\boxed{\psi_\mathbf{p}(\mathbf{r},t) = \frac{1}{\mathcal{V}^{1/2}}\exp\Big\{\frac{i}{\hbar}\Big[\mathbf{p}\cdot\mathbf{r} - (E_\mathbf{p}+U_\mathrm{p})\,t + \mathbf{p}\cdot\mathbf{a}_0\,(1-\cos\omega t) + \frac{U_\mathrm{p}}{2\omega}\sin 2\omega t\Big]\Big\},} \tag{4.5}$$

which is Eq. (1) [3] with the integral in its phase carried out for the field of Eq. (3.1), apart from the normalization.
(3) The phase $\mathbf{p}\cdot\mathbf{r} + \mathbf{p}\cdot\mathbf{a}_0(1-\cos\omega t) = \mathbf{p}\cdot[\mathbf{r} - \boldsymbol{\alpha}(t) + \boldsymbol{\alpha}(0)]$ shows that the Volkov wave is the plane wave of the oscillating frame of Section 3.3, carried along with the quiver motion. The mean energy is $E_\mathbf{p} + U_\mathrm{p}$, and U_p is the same for all states, so it drops out of every energy difference.

4.2 Orthonormality and completeness

At equal times the time-dependent phases are the same for the two factors of a product with equal momenta, so

$$\int \psi^*_{\mathbf{p}'}(\mathbf{r},t)\psi_\mathbf{p}(\mathbf{r},t)\mathrm{d}^3r = \delta_{\mathbf{p}\mathbf{p}'}, \qquad \sum_\mathbf{p}\psi_\mathbf{p}(\mathbf{r},t)\psi^*_\mathbf{p}(\mathbf{r}',t) = \frac{1}{\mathcal{V}}\sum_\mathbf{p} e^{i\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}')/\hbar} = \delta(\mathbf{r}-\mathbf{r}'). \tag{4.6}$$

The Volkov states at time t are a complete orthonormal set, and they can be used as the basis of the many-electron field operator in Section 6.

4.3 The Jacobi–Anger expansion and three sum rules

(1) The generating function of the Bessel functions is $e^{\frac{1}{2}z(\tau-\tau^{-1})} = \sum_\ell \tau^\ell J_\ell(z)$, Eq. (10.12.1) [11]. With $\tau = ie^{i\theta}$, $\tau - \tau^{-1} = ie^{i\theta} + ie^{-i\theta} = 2i\cos\theta$, and

$$\boxed{e^{iz\cos\theta} = \sum_{\ell=-\infty}^{\infty} i^\ell J_\ell(z)\, e^{i\ell\theta}.} \tag{4.7}$$

(2) Parseval's theorem for the Fourier series in θ and for the series of its derivative $\partial_\theta e^{iz\cos\theta} = -iz\sin\theta\, e^{iz\cos\theta} = \sum_\ell i^\ell (i\ell) J_\ell e^{i\ell\theta}$ gives

$$\sum_\ell J_\ell^2(z) = \frac{1}{2\pi}\int_0^{2\pi} \left|e^{iz\cos\theta}\right|^2 \mathrm{d}\theta = 1, \qquad \sum_\ell \ell^2 J_\ell^2(z) = \frac{1}{2\pi}\int_0^{2\pi} z^2\sin^2\theta\,\mathrm{d}\theta = \frac{z^2}{2}. \tag{4.8}$$

Since $J_{-\ell} = (-1)^\ell J_\ell$, the squares are even in ℓ and the first moment vanishes. Altogether,

$$\boxed{\sum_\ell J_\ell^2(z) = 1, \qquad \sum_\ell \ell\, J_\ell^2(z) = 0, \qquad \sum_\ell \ell^2 J_\ell^2(z) = \frac{z^2}{2}, \qquad J_{-\ell}^2 = J_\ell^2.} \tag{4.9}$$

(3) For small arguments $J_\ell(z) = (\pm z/2)^{|\ell|}/|\ell|! + O(z^{|\ell|+2})$, so that

$$J_0^2(z) = 1 - \frac{z^2}{2} + O(z^4), \qquad J_{\pm1}^2(z) = \frac{z^2}{4} + O(z^4), \qquad J_{\pm\ell}^2(z) = O(z^{2|\ell|}). \tag{4.10}$$

(4) Multiplying Eq. (4.7) by the complex conjugate of the same expansion at $\theta + s$ and averaging over θ gives the generating function of the squares. With $\cos\theta - \cos(\theta+s) = 2\sin(s/2)\sin(\theta + s/2)$ and $(2\pi)^{-1}\int_0^{2\pi} e^{ix\sin\varphi}\mathrm{d}\varphi = J_0(x)$,

$$\boxed{\sum_\ell J_\ell^2(z)\, e^{-i\ell s} = \frac{1}{2\pi}\int_0^{2\pi} e^{iz[\cos\theta - \cos(\theta+s)]}\mathrm{d}\theta = J_0\!\left(2z\sin\frac{s}{2}\right).} \tag{4.11}$$

This identity gives the classical limit in Section 5.6.

5 Multiphoton transition rates in Born approximation

5.1 The transition amplitude

(1) Expand the state of one electron in the Volkov states, $\psi(t) = \sum_{\mathbf{p}} c_{\mathbf{p}}(t)\psi_{\mathbf{p}}(t)$. Since each $\psi_{\mathbf{p}}$ solves the equation without ions, the Schrödinger equation with H of Eq. (3.5) reduces to

$$i\hbar\,\dot{c}_{\mathbf{p}'} = \sum_{\mathbf{p}} V_{\mathbf{p}'\mathbf{p}}(t)\, c_{\mathbf{p}}, \qquad V_{\mathbf{p}'\mathbf{p}}(t) = \int \psi^*_{\mathbf{p}'}(\mathbf{r},t) V(\mathbf{r}) \psi_{\mathbf{p}}(\mathbf{r},t)\mathrm{d}^3 r. \tag{5.1}$$

To first order in V, starting from $c_{\mathbf{p}}(0) = \delta_{\mathbf{p}\mathbf{p}_i}$,

$$c_{\mathbf{p}'}(t) = -\frac{i}{\hbar}\int_0^t V_{\mathbf{p}'\mathbf{p}_i}(t')\mathrm{d}t'. \tag{5.2}$$

(2) In the matrix element the factor $\mathcal{V}^{-1}e^{-i(\mathbf{p}'-\mathbf{p})\cdot\mathbf{r}/\hbar}$ multiplies $V(\mathbf{r})$, and with $\hbar\mathbf{k}=\mathbf{p}'-\mathbf{p}$ the spatial integral is $\tilde{V}(\mathbf{k})/\mathcal{V}$ of Eq. (3.8). The time-dependent phases of Eq. (4.5) combine to

$$\begin{aligned}\hbar(\Phi_{\mathbf{p}'}-\Phi_{\mathbf{p}}) &= (E_{\mathbf{p}'}-E_{\mathbf{p}})\,t + \cancel{(U_\mathrm{p}-U_\mathrm{p})\,t} - (\mathbf{p}'-\mathbf{p})\cdot\mathbf{a}_0\,(1-\cos\omega t) - \cancel{\Big(\frac{U_\mathrm{p}}{2\omega}-\frac{U_\mathrm{p}}{2\omega}\Big)\sin 2\omega t} \\ &= (E_{\mathbf{p}'}-E_{\mathbf{p}})\,t - \hbar\mathbf{k}\cdot\mathbf{a}_0\,(1-\cos\omega t),\end{aligned} \tag{5.3}$$

in which the terms with U_p cancel because the ponderomotive energy is the same for every state. With $z\equiv\mathbf{k}\cdot\mathbf{a}_0$ and Eq. (4.7),

$$\boxed{V_{\mathbf{p}'\mathbf{p}}(t) = \frac{\tilde{V}(\mathbf{k})}{\mathcal{V}}\,e^{-iz}\sum_{\ell=-\infty}^{\infty} i^\ell J_\ell(z)\,\exp\Big[\frac{i}{\hbar}\big(E_{\mathbf{p}'}-E_{\mathbf{p}}+\ell\hbar\omega\big)t\Big], \qquad \hbar\mathbf{k}=\mathbf{p}'-\mathbf{p}, \quad z=\mathbf{k}\cdot\mathbf{a}_0.} \tag{5.4}$$

The matrix element is a sum of harmonics of the laser frequency, and the ℓth harmonic carries the weight $J_\ell(z)$.

5.2 The long-time limit

(1) Insert Eq. (5.4) into Eq. (5.2),

$$c_{\mathbf{p}'}(t) = -\frac{i}{\hbar}\,\frac{\tilde{V}(\mathbf{k})}{\mathcal{V}}\,e^{-iz}\sum_\ell i^\ell J_\ell(z)\,I_\ell(t), \qquad I_\ell(t)=\int_0^t e^{i\Omega_\ell t'}\,\mathrm{d}t' = e^{i\Omega_\ell t/2}\,\frac{\sin(\Omega_\ell t/2)}{\Omega_\ell/2}, \tag{5.5}$$

with $\hbar\Omega_\ell = E_{\mathbf{p}'}-E_{\mathbf{p}}+\ell\hbar\omega$. The probability is

$$|c_{\mathbf{p}'}(t)|^2 = \frac{|\tilde{V}(\mathbf{k})|^2}{\hbar^2\mathcal{V}^2}\Big[\sum_\ell J_\ell^2(z)\,|I_\ell(t)|^2 + \sum_{\ell\neq\ell'} i^{\ell-\ell'}J_\ell(z)J_{\ell'}(z)\,I_\ell(t)I_{\ell'}^*(t)\Big]. \tag{5.6}$$

(2) The diagonal terms grow linearly in time. With $x=\Omega t/2$, $\int\sin^2(\Omega t/2)/(\Omega/2)^2\,\mathrm{d}\Omega = 2t\int\sin^2 x/x^2\,\mathrm{d}x = 2\pi t$, and the function concentrates at $\Omega=0$ as t grows, so that

$$|I_\ell(t)|^2 = \frac{\sin^2(\Omega_\ell t/2)}{(\Omega_\ell/2)^2} \;\longrightarrow\; 2\pi t\,\delta(\Omega_\ell) = 2\pi\hbar t\,\delta(\hbar\Omega_\ell). \tag{5.7}$$

(3) The cross terms do not grow. The peaks of $|I_\ell|$ and $|I_{\ell'}|$ sit at energies that differ by $(\ell-\ell')\hbar\omega$, so an integral of $I_\ell I_{\ell'}^*$ over the final energies stays bounded as $t\to\infty$. They describe the modulation of the scattering within a period of the field, which produces harmonics of the current [9] and averages out of the absorbed power.
(4) The transition rate $W=\mathrm{d}|c_{\mathbf{p}'}|^2/\mathrm{d}t$ is then

$$W_{\mathbf{p}\to\mathbf{p}'} = \frac{2\pi}{\hbar}\,\frac{|\tilde{V}(\mathbf{k})|^2}{\mathcal{V}^2}\sum_\ell J_\ell^2(z)\,\delta\big(E_{\mathbf{p}'}-E_{\mathbf{p}}+\ell\hbar\omega\big). \tag{5.8}$$

Replacing ℓ by $-\ell$, which leaves J_ℓ^2 unchanged by Eq. (4.9), puts it in the form in which $\ell>0$ counts absorbed photons,

$$\boxed{W_{\mathbf{p}\to\mathbf{p}'} = \frac{2\pi}{\hbar}\,\frac{|\tilde{V}(\mathbf{k})|^2}{\mathcal{V}^2}\sum_{\ell=-\infty}^{\infty} J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\,\delta\big(E_{\mathbf{p}'}-E_{\mathbf{p}}-\ell\hbar\omega\big), \qquad \hbar\mathbf{k}=\mathbf{p}'-\mathbf{p}.} \tag{5.9}$$

The rate is symmetric, $W_{\mathbf{p}\to\mathbf{p}'}=W_{\mathbf{p}'\to\mathbf{p}}$, because the term ℓ of one equals the term $-\ell$ of the other and $|\tilde{V}(-\mathbf{k})|=|\tilde{V}(\mathbf{k})|$.

5.3 The average over the ion positions

(1) By Eq. (3.8), $|\tilde{V}(\mathbf{k})|^2 = |\tilde{v}(\mathbf{k})|^2 |\rho_i(\mathbf{k})|^2$ with

$$|\rho_i(\mathbf{k})|^2 = \sum_{j,j'} e^{-i\mathbf{k}\cdot(\mathbf{R}_j - \mathbf{R}_{j'})} = N_i + \sum_{j \neq j'} e^{-i\mathbf{k}\cdot(\mathbf{R}_j - \mathbf{R}_{j'})}. \tag{5.10}$$

(2) For ions placed independently and uniformly in the box, $\langle e^{-i\mathbf{k}\cdot(\mathbf{R}_j - \mathbf{R}_{j'})} \rangle = |\langle e^{-i\mathbf{k}\cdot\mathbf{R}} \rangle|^2 = \delta_{\mathbf{k}0}$ for $j \neq j'$, so $\langle |\rho_i(\mathbf{k})|^2 \rangle = N_i$ for $\mathbf{k} \neq 0$. For correlated ions the average defines the static structure factor, $\langle |\rho_i(\mathbf{k})|^2 \rangle = N_i S(\mathbf{k})$, and $S = 1$ for independent ions.
(3) The averaged rate is

$$\boxed{W_{\mathbf{p}\to\mathbf{p}'} = \frac{2\pi}{\hbar} \frac{n_i}{\mathcal{V}} |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) \sum_{\ell} J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\, \delta\big(E_{\mathbf{p}'} - E_{\mathbf{p}} - \ell\hbar\omega\big).} \tag{5.11}$$

5.4 The cross section of Bunkin and Fedorov

(1) The rate of transitions of one electron into the final directions $\mathrm{d}\Omega'$ is the sum of Eq. (5.11) over the final states in that cone. With $\sum_{\mathbf{p}'} \to \mathcal{V} \int p'^2 \mathrm{d}p'\, \mathrm{d}\Omega'/(2\pi\hbar)^3$ and $\mathrm{d}E_{\mathbf{p}'} = p'\mathrm{d}p'/m$, the energy delta function sets $p' = p'_\ell \equiv (p^2 + 2m\ell\hbar\omega)^{1/2}$ and contributes mp'_ℓ,

$$\mathrm{d}\Gamma_\ell = \frac{2\pi}{\hbar}\, n_i |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\, \frac{mp'_\ell}{(2\pi\hbar)^3}\, \mathrm{d}\Omega'. \tag{5.12}$$

(2) Dividing by the flux $n_i v = n_i p/m$ of ions seen by the electron gives the cross section for the absorption of ℓ photons,

$$\boxed{\begin{aligned} \frac{\mathrm{d}\sigma_\ell}{\mathrm{d}\Omega'} &= \frac{m^2}{4\pi^2\hbar^4} \frac{p'_\ell}{p} |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k})\, J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) = \frac{p'_\ell}{p}\, J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\, \frac{\mathrm{d}\sigma_B}{\mathrm{d}\Omega}(\mathbf{k}), \\ \frac{\mathrm{d}\sigma_B}{\mathrm{d}\Omega} &= \Big(\frac{m}{2\pi\hbar^2}\Big)^2 |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}). \end{aligned}} \tag{5.13}$$

Here $\mathrm{d}\sigma_B/\mathrm{d}\Omega$ is the field-free Born cross section at the momentum transfer $\hbar\mathbf{k}$. Equation (5.13) is Eq. (3) [3], which normalizes the cross section by the photon flux $I/\hbar\omega$ with $I = cE_0^2/8\pi$ and writes the argument of the Bessel function as $\gamma[\cos\theta_0(1 - \lambda\cos\theta) - \lambda\sin\theta_0\sin\theta\cos\varphi]$, which is $-\mathbf{k}\cdot\mathbf{a}_0$ in the present notation.
(3) For a bare Coulomb ion and $S = 1$, with $\hbar k = 2p\sin(\theta/2)$ for elastic scattering,

$$\frac{\mathrm{d}\sigma_B}{\mathrm{d}\Omega} = \frac{4m^2Z^2e^4}{\hbar^4k^4} = \Big(\frac{Ze^2}{2mv^2}\Big)^2 \frac{1}{\sin^4(\theta/2)}, \tag{5.14}$$

the Rutherford cross section. At $E_0 = 0$ only $\ell = 0$ survives, with $J_0(0) = 1$, and Eq. (5.13) reduces to it.
(4) When $\ell\hbar\omega \ll E_{\mathbf{p}}$ the factor p'_ℓ/p is close to one, and the sum rule $\sum_\ell J_\ell^2 = 1$ shows that the field does not change the total scattering. It distributes each scattering event over the energies $E_{\mathbf{p}} + \ell\hbar\omega$.

5.5 Weak fields and one photon

(1) For $|\mathbf{k}\cdot\mathbf{a}_0| \ll 1$, Eq. (4.10) keeps $\ell = 0, \pm 1$. With Eq. (3.7) at $\kappa = 0$ and $S = 1$, the one-photon cross sections are

$$\frac{\mathrm{d}\sigma_{\pm 1}}{\mathrm{d}\Omega'} = \frac{m^2}{4\pi^2\hbar^4}\,\frac{p'_\pm}{p}\,\frac{16\pi^2 Z^2 e^4}{k^4}\,\frac{(\mathbf{k}\cdot\mathbf{a}_0)^2}{4} = \frac{m^2 Z^2 e^4 a_0^2}{\hbar^2}\,\frac{p'_\pm}{p}\,\frac{[(\mathbf{p}'-\mathbf{p})\cdot\hat{\mathbf{e}}]^2}{|\mathbf{p}'-\mathbf{p}|^4}, \tag{5.15}$$

with $p'_\pm = (p^2 \pm 2m\hbar\omega)^{1/2}$. The upper sign is absorption and the lower one stimulated emission, which exists only for $E_\mathbf{p} > \hbar\omega$.

(2) Let the incident momentum make the angle θ_0 with $\hat{\mathbf{e}}$. Put $\mathbf{p}$ along the z axis, $\hat{\mathbf{e}} = (\sin\theta_0, 0, \cos\theta_0)$ and $\mathbf{p}' = p'(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. The average over φ is

$$\big\langle[(\mathbf{p}'-\mathbf{p})\cdot\hat{\mathbf{e}}]^2\big\rangle_\varphi = \tfrac{1}{2}p'^2\sin^2\theta\,\sin^2\theta_0 + (p'\cos\theta - p)^2\cos^2\theta_0. \tag{5.16}$$

With $D = p^2 + p'^2 - 2pp'c$, $c = \cos\theta$, the remaining integrals are elementary. Using $b = 2pp'$, $a = p^2 + p'^2$, $\Delta = p^2 - p'^2$ and $L = \ln[(p+p')/|p-p'|]$,

$$\int_{-1}^{1}\frac{\mathrm{d}c}{D^2} = \frac{2}{\Delta^2}, \qquad \int_{-1}^{1}\frac{c\,\mathrm{d}c}{D^2} = \frac{2a}{b\Delta^2} - \frac{2L}{b^2}, \qquad \int_{-1}^{1}\frac{c^2\mathrm{d}c}{D^2} = \frac{2a^2}{b^2\Delta^2} - \frac{4aL}{b^3} + \frac{2}{b^2}, \tag{5.17}$$

from which, using $b^2 - a^2 = -\Delta^2$ and $(a - 2p^2)^2 = \Delta^2$,

$$\int_{-1}^{1}\frac{(1-c^2)\,\mathrm{d}c}{D^2} = \frac{(p^2+p'^2)L}{2p^3p'^3} - \frac{1}{p^2p'^2}, \qquad \int_{-1}^{1}\frac{(p'c-p)^2\mathrm{d}c}{D^2} = \frac{1}{p^2} + \frac{(p^2-p'^2)L}{2p^3p'}. \tag{5.18}$$

(3) Collecting,

$$\boxed{\sigma_{\pm1}(\theta_0) = \frac{2\pi m^2 Z^2 e^4 a_0^2}{\hbar^2 p^2}\,\frac{p'_\pm}{p}\Big\{\sin^2\theta_0\Big[\frac{(p^2+p'^2_\pm)L_\pm}{4pp'_\pm} - \frac{1}{2}\Big] + \cos^2\theta_0\Big[1 + \frac{(p^2-p'^2_\pm)L_\pm}{2pp'_\pm}\Big]\Big\},} \tag{5.19}$$

with $L_\pm = \ln[(p+p'_\pm)/|p-p'_\pm|]$.

(4) The average over the orientation of $\mathbf{p}$ uses $\langle\sin^2\theta_0\rangle = 2/3$ and $\langle\cos^2\theta_0\rangle = 1/3$. The terms without L cancel and

$$\boxed{\bar\sigma_{\pm1}(p) = \frac{2\pi}{3}\,\frac{m^2Z^2e^4a_0^2}{\hbar^2p^2}\,L_\pm, \qquad L_\pm = \ln\frac{p+p'_\pm}{|p-p'_\pm|}.} \tag{5.20}$$

(5) Bunkin and Fedorov expand Eq. (5.19) for $\xi = \hbar\omega/mv^2 \ll 1$, in their normalization per photon flux. Multiplying by $n_i v\hbar\omega/I$ with $I = cE_0^2/8\pi$ and $a_0^2 = e^2E_0^2/m^2\omega^4$ turns the prefactor of Eq. (5.19) into $16\pi^2 n_i Z^2 e^6/(m^2 vc\hbar\omega^3)$. With $p'_\pm = p(1\pm2\xi)^{1/2}$, $L_\pm = \ln(2/\xi) \pm \xi + O(\xi^2)$ and the curly bracket to first order in ξ, the result is

$$\sigma^{\mathrm{BF}}_{a,e} = \frac{8\pi^2Z^2e^6n_i}{m^2vc\hbar\omega^3}\Big[(3\cos^2\theta_0 - 1) + (1-\cos^2\theta_0)\ln\frac{2}{\xi} \mp (3\cos^2\theta_0 - 1)\,\xi\ln\frac{2}{\xi} \pm 2\xi\cos^2\theta_0\Big], \tag{5.21}$$

which is Eq. (10) [3] term by term, with the upper sign for absorption. The net emission $\sigma_e - \sigma_a$ is proportional to $(3\cos^2\theta_0 - 1)\ln(2/\xi) - 2\cos^2\theta_0$. It is positive for electrons moving along the field, which then amplify it, as Marcuse first found [13]. Averaged over directions it is $-2/3$ times the prefactor, and the plasma absorbs.

5.6 The classical limit of the photon-number distribution

(1) Let $\hbar\omega \to 0$ at a fixed quiver velocity, so that $z = \mathbf{k}\cdot\mathbf{a}_0 \to \infty$ while $Z_E \equiv z\hbar\omega = \hbar\mathbf{k}\cdot\hat{\mathbf{e}}\,v_0$ stays fixed. Take a smooth test function $\phi(\epsilon)$ of the energy transfer with Fourier representation $\phi(\epsilon) = \int \hat{\phi}(\tau)e^{-i\epsilon\tau}\mathrm{d}\tau$. By Eq. (4.11) with $s = \hbar\omega\tau$,

$$\sum_\ell J_\ell^2(z)\,\phi(\ell\hbar\omega) = \int \hat{\phi}(\tau)\,J_0\Big(2z\sin\frac{\hbar\omega\tau}{2}\Big)\mathrm{d}\tau \;\longrightarrow\; \int \hat{\phi}(\tau)\,J_0(Z_E\tau)\,\mathrm{d}\tau. \tag{5.22}$$

(2) With $J_0(x) = (2\pi)^{-1}\int_0^{2\pi} e^{-ix\cos\theta}\mathrm{d}\theta$ the limit is $(2\pi)^{-1}\int_0^{2\pi}\phi(Z_E\cos\theta)\,\mathrm{d}\theta$. In distributional form,

$$\boxed{\sum_\ell J_\ell^2(z)\,\delta(\epsilon-\ell\hbar\omega) \;\longrightarrow\; \frac{1}{2\pi}\int_0^{2\pi}\delta(\epsilon - Z_E\cos\theta)\,\mathrm{d}\theta = \frac{1}{\pi}\,\frac{\Theta(Z_E^2-\epsilon^2)}{(Z_E^2-\epsilon^2)^{1/2}}, \qquad Z_E = \hbar|\mathbf{k}\cdot\hat{\mathbf{e}}|\,v_0.} \tag{5.23}$$

(3) This is the classical picture. The ion is at rest in the laboratory, so a collision that is short compared with the period conserves the kinetic energy $m|\mathbf{v}|^2/2$ of the electron while its momentum changes by $\hbar\mathbf{k}$. The kinetic momentum is $m\mathbf{v} = \mathbf{p} + m\mathbf{v}_{\rm osc}$, so the drift energy $E_{\mathbf{p}} = |m\mathbf{v} - m\mathbf{v}_{\rm osc}|^2/2m$ changes by $\Delta\epsilon = -\hbar\mathbf{k}\cdot\mathbf{v}_{\rm osc}(t_c) = \hbar(\mathbf{k}\cdot\hat{\mathbf{e}})v_0\sin\omega t_c$, where t_c is the time of the collision. A uniformly distributed phase ωt_c gives the arcsine density of Eq. (5.23).

Remarks:
(1) The Born approximation for a Coulomb ion needs $\eta = Ze^2/\hbar v \ll 1$, Eq. (5) [3]. Since $\eta = r_0/\lambda\!\!\!^{-}$ with $r_0 = Ze^2/mv^2$ and $\lambda\!\!\!^{-} = \hbar/mv$, it is the condition that the de Broglie wavelength exceeds the distance of closest approach. It is the same condition that selects the quantum lower cutoff of the Coulomb logarithm in [1, Eq. (10.136)] and the Bethe rather than the Bohr stopping number in [1, Eq. (11.74)]. At the thermal velocity $\eta = Z(2\,\mathrm{Ry}/k_BT)^{1/2}$, with $2\,\mathrm{Ry} = 27.2$ eV.
(2) At first order in E_0 the amplitude of Eq. (5.4) with $\ell = 1$ is that of second-order perturbation theory, with one photon vertex $(e/mc)\mathbf{A}\cdot\hat{\mathbf{p}}$ and one ion vertex in both orders. The photon vertex is diagonal in momentum, so the two orders have the intermediate states $\mathbf{p}$ and $\mathbf{p}'$, the energy denominators $\hbar\omega$ and $-\hbar\omega$, and the amplitude $(e/mc\hbar\omega)\mathbf{A}_1\cdot(\mathbf{p}-\mathbf{p}')\tilde{V}(\mathbf{k})/\mathcal{V}$, where $\mathbf{A}_1$ is the amplitude of $e^{-i\omega t}$ in $\mathbf{A}$, of magnitude $cE_0/2\omega$. Its magnitude is $\frac{1}{2}|\mathbf{k}\cdot\mathbf{a}_0||\tilde{V}|/\mathcal{V}$, which is $J_1(\mathbf{k}\cdot\mathbf{a}_0)|\tilde{V}|/\mathcal{V}$ to this order. The term $e^2A^2/2mc^2$ only shifts all energies by $U_{\rm p}$ and does not contribute.
(3) The weak-field cross sections of Eq. (5.20) are what the kinetic equation of Section 12 needs for an isotropic distribution.

6 The kinetic equation

This section derives the equation of motion of the occupation numbers with the operator method that Section 2 of the quantum Boltzmann equation paper [4] uses for a quartic interaction. The electrons are taken as independent here. Their interaction enters in Section 10.

6.1 The field operator in the Volkov basis

(1) Expand the electron field operator of spin σ in the Volkov states, which are complete at every time by Eq. (4.6),

$$\hat{\Psi}_\sigma(\mathbf{r},t) = \sum_{\mathbf{p}} \psi_{\mathbf{p}}(\mathbf{r},t)\,\hat{b}_{\mathbf{p}\sigma}, \qquad \hat{b}_{\mathbf{p}\sigma}\hat{b}^\dagger_{\mathbf{p}'\sigma'} - \eta_s\,\hat{b}^\dagger_{\mathbf{p}'\sigma'}\hat{b}_{\mathbf{p}\sigma} = \delta_{\mathbf{p}\mathbf{p}'}\delta_{\sigma\sigma'}, \tag{6.1}$$

with $\eta_s = -1$ for electrons. The value $\eta_s = +1$ for bosons is kept to show that the result does not depend on it.
(2) In the interaction picture with respect to the motion in the laser field, the time dependence of the free motion is carried by the mode functions $\psi_{\mathbf{p}}$, the operators $\hat{b}_{\mathbf{p}\sigma}$ are constant in the absence of ions, and the ion potential is the one-body operator

$$\hat{H}_V(t) = \sum_\sigma \int \hat{\Psi}^\dagger_\sigma V \hat{\Psi}_\sigma \, \mathrm{d}^3 r = \sum_\sigma \sum_{\mathbf{p},\mathbf{p}'} V_{\mathbf{p}'\mathbf{p}}(t)\, \hat{b}^\dagger_{\mathbf{p}'\sigma} \hat{b}_{\mathbf{p}\sigma}, \tag{6.2}$$

with the matrix elements of Eq. (5.4). The potential does not act on the spin, and the spin label is dropped from here on.

6.2 The exact equation of motion of the one-body density matrix

(1) The commutator of two bilinears is the same for both statistics. For fermions,

$$\hat{b}^\dagger_a \hat{b}_b \hat{b}^\dagger_c \hat{b}_d = \delta_{bc} \hat{b}^\dagger_a \hat{b}_d - \hat{b}^\dagger_a \hat{b}^\dagger_c \hat{b}_b \hat{b}_d, \qquad \hat{b}^\dagger_c \hat{b}_d \hat{b}^\dagger_a \hat{b}_b = \delta_{ad} \hat{b}^\dagger_c \hat{b}_b - \hat{b}^\dagger_c \hat{b}^\dagger_a \hat{b}_d \hat{b}_b, \tag{6.3}$$

and the quartic terms are equal because $\hat{b}^\dagger_c \hat{b}^\dagger_a \hat{b}_d \hat{b}_b = (-\hat{b}^\dagger_a \hat{b}^\dagger_c)(-\hat{b}_b \hat{b}_d)$. For bosons both signs change and the quartic terms are again equal. Hence

$$\big[\hat{b}^\dagger_a \hat{b}_b, \hat{b}^\dagger_c \hat{b}_d\big] = \delta_{bc}\, \hat{b}^\dagger_a \hat{b}_d - \delta_{ad}\, \hat{b}^\dagger_c \hat{b}_b. \tag{6.4}$$

(2) The Heisenberg equation for $\hat{b}^\dagger_k \hat{b}_{k'}$, with k, k' short for momenta, is then

$$\frac{\mathrm{d}}{\mathrm{d}t}\big(\hat{b}^\dagger_k \hat{b}_{k'}\big) = \frac{i}{\hbar}\big[\hat{H}_V, \hat{b}^\dagger_k \hat{b}_{k'}\big] = \frac{i}{\hbar} \sum_{c,d} V_{cd}\big(\delta_{dk} \hat{b}^\dagger_c \hat{b}_{k'} - \delta_{ck'} \hat{b}^\dagger_k \hat{b}_d\big) = \frac{i}{\hbar} \sum_c \big(V_{ck}\, \hat{b}^\dagger_c \hat{b}_{k'} - V_{k'c}\, \hat{b}^\dagger_k \hat{b}_c\big). \tag{6.5}$$

With the one-body density matrix $\rho_{ab} = \langle \hat{b}^\dagger_a \hat{b}_b \rangle$,

$$\boxed{\frac{\mathrm{d}\rho_{kk'}}{\mathrm{d}t} = \frac{i}{\hbar} \sum_c \big[V_{ck}(t)\, \rho_{ck'} - V_{k'c}(t)\, \rho_{kc}\big].} \tag{6.6}$$

(3) Equation (6.6) is exact and closed. In the quantum Boltzmann equation paper the quartic interaction makes the right side of Eq. (4) [4] a product of four operators, whose average at second order needs the six-field correlator and Wick's theorem. Here the ion potential is a one-body operator, the right side is again bilinear, and no factorization is needed.

6.3 Second order and the Markov limit

(1) Let the initial state be homogeneous, $\rho_{ab}(0) = \delta_{ab} f_a$. To first order in V, Eq. (6.6) has $\sum_c V_{ca}\rho_{cb} = V_{ba} f_b$ and $\sum_c V_{bc}\rho_{ac} = V_{ba} f_a$, so

$$\rho^{(1)}_{ab}(t) = \frac{i}{\hbar}\,(f_b - f_a) \int_0^t V_{ba}(t')\, \mathrm{d}t'. \tag{6.7}$$

The diagonal elements do not change at first order, because $f_k - f_k = 0$. This is the vanishing of the mean-field term that the quantum Boltzmann equation paper finds for $\mathbf{k} = \mathbf{k}'$ in its Eq. (5) [4].

(2) For the diagonal element at second order, insert Eq. (6.7) into Eq. (6.6) with $k' = k$. Since $\rho^{(1)}_{kc} = \rho^{(1)*}_{ck}$ and $V_{kc} = V^*_{ck}$, the two terms are $\zeta_c - \zeta_c^* = 2i\,\mathrm{Im}\,\zeta_c$ with $\zeta_c = V_{ck}(t)\rho^{(1)}_{ck}(t) = (i/\hbar)(f_k - f_c)V_{ck}(t)\int_0^t V_{kc}(t')\mathrm{d}t'$, and

$$\frac{\mathrm{d}f_k}{\mathrm{d}t} = -\frac{2}{\hbar}\sum_c \mathrm{Im}\,\zeta_c = \frac{2}{\hbar^2}\sum_c (f_c - f_k)\;\mathrm{Re}\left[V^*_{kc}(t)\int_0^t V_{kc}(t')\,\mathrm{d}t'\right]. \tag{6.8}$$

Since $\mathrm{Re}[V^*(t)\int_0^t V\mathrm{d}t'] = \frac{1}{2}\mathrm{d}|\int_0^t V\mathrm{d}t'|^2/\mathrm{d}t$, the bracket is $\hbar^2/2$ times the rate of growth of the transition probability of Eq. (5.2).
(3) Insert Eq. (5.4). The phases e^{-iz} cancel between V^*_{kc} and V_{kc}, and with $\hbar\Omega_\ell = E_k - E_c + \ell\hbar\omega$ and $\tau = t - t'$,

$$\mathrm{Re}\left[V^*_{kc}(t)\int_0^t V_{kc}(t')\mathrm{d}t'\right] = \frac{|\tilde{V}|^2}{\mathcal{V}^2}\sum_{\ell,\ell'}\mathrm{Re}\left[i^{\ell'-\ell}J_\ell J_{\ell'}\,e^{i(\ell'-\ell)\omega t}\int_0^t e^{-i\Omega_{\ell'}\tau}\mathrm{d}\tau\right]. \tag{6.9}$$

The terms with $\ell' \neq \ell$ oscillate at multiples of ω and average to zero over a period. The terms with $\ell' = \ell$ contain

$$\mathrm{Re}\int_0^t e^{-i\Omega\tau}\mathrm{d}\tau = \frac{\sin\Omega t}{\Omega} \;\longrightarrow\; \pi\,\delta(\Omega) \qquad (t\to\infty). \tag{6.10}$$

The one-sided time integral gives $\pi\delta$, and the factor 2 of Eq. (6.8), which comes from the two Hermitian-conjugate terms, completes the 2π of the golden rule. This is the step at which the published Eqs. (5)–(7) [4] carry twice the coefficient, as the erratum [5] corrects.
(4) With $\delta(\Omega) = \hbar\,\delta(\hbar\Omega)$ the result is $\mathrm{d}f_k/\mathrm{d}t = \sum_c W_{c\to k}(f_c - f_k)$ with the rate of Eq. (5.9), and after the average over the ion positions with the rate of Eq. (5.11),

$$\boxed{\frac{\partial f_{\mathbf{p}}}{\partial t} = \sum_{\mathbf{p}'} W_{\mathbf{p}'\to\mathbf{p}}\,(f_{\mathbf{p}'} - f_{\mathbf{p}}), \qquad W_{\mathbf{p}'\to\mathbf{p}} = W_{\mathbf{p}\to\mathbf{p}'}.} \tag{6.11}$$

This is the kinetic equation of inverse bremsstrahlung, a master equation in which each pair of states is connected by the multiphoton rates of Bunkin and Fedorov.

6.4 Why there is no Pauli blocking

The Uehling–Uhlenbeck form of the collision integral, Eq. (7) [4], carries the factors $1 + \eta_s n$ of the final states. For scattering on a static potential the gain and the loss term of one pair of states combine as

$$f_{\mathbf{p}'}(1 + \eta_s f_{\mathbf{p}}) - f_{\mathbf{p}}(1 + \eta_s f_{\mathbf{p}'}) = f_{\mathbf{p}'} - f_{\mathbf{p}}, \tag{6.12}$$

because the quadratic terms are equal. The exact one-body equation of motion never produced them. Pauli blocking does not suppress inverse bremsstrahlung on static ions, but the Fermi–Dirac distribution enters through f itself, Section 8.2.

6.5 Conservation, equilibrium and the heating rate

(1) Summing Eq. (6.11) over $\mathbf{p}$ and using the symmetry of W shows that the number of electrons is conserved. Momentum is not conserved, because the ions absorb it.
(2) Without the field only $\ell = 0$ survives, W contains $\delta(E_{\mathbf{p}'} - E_{\mathbf{p}})$, and every distribution that depends on the energy alone is stationary. The field drives the electrons through the terms with $\ell \neq 0$.

(3) The kinetic energy of the drift motion per unit volume, counting both spin states, is $u_e = (2/\mathcal{V})\sum_{\mathbf{p}} E_{\mathbf{p}} f_{\mathbf{p}}$. The quiver energy U_{p} per electron is constant and absorbs nothing on average. With Eq. (6.11), and then with the labels $\mathbf{p}, \mathbf{p}'$ exchanged and the two forms averaged,

$$P = \frac{\mathrm{d}u_e}{\mathrm{d}t} = \frac{2}{\mathcal{V}} \sum_{\mathbf{p},\mathbf{p}'} E_{\mathbf{p}} W_{\mathbf{p}'\to\mathbf{p}} (f_{\mathbf{p}'} - f_{\mathbf{p}}) = \frac{1}{\mathcal{V}} \sum_{\mathbf{p},\mathbf{p}'} \big(E_{\mathbf{p}} - E_{\mathbf{p}'}\big) W_{\mathbf{p}'\to\mathbf{p}} (f_{\mathbf{p}'} - f_{\mathbf{p}}). \tag{6.13}$$

(4) The term ℓ of the rate fixes $E_{\mathbf{p}} - E_{\mathbf{p}'} = \ell\hbar\omega$. Pairing the term $-\ell$ with the term ℓ after exchanging $\mathbf{p}$ and $\mathbf{p}'$ in it, which uses $J_{-\ell}^2 = J_\ell^2$, gives

$$\boxed{\begin{aligned} P &= \frac{2}{\mathcal{V}} \sum_{\ell=1}^{\infty} \ell\hbar\omega \sum_{\mathbf{p},\mathbf{p}'} w_\ell(\mathbf{p}' \to \mathbf{p}) \, (f_{\mathbf{p}'} - f_{\mathbf{p}}), \\ w_\ell(\mathbf{p}' \to \mathbf{p}) &= \frac{2\pi}{\hbar} \frac{n_i}{\mathcal{V}} |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) \, \delta(E_{\mathbf{p}} - E_{\mathbf{p}'} - \ell\hbar\omega), \end{aligned}} \tag{6.14}$$

with $\hbar\mathbf{k} = \mathbf{p} - \mathbf{p}'$. Each term is a transition from the lower energy $E_{\mathbf{p}'}$ to the higher one $E_{\mathbf{p}} = E_{\mathbf{p}'} + \ell\hbar\omega$, absorption minus stimulated emission.
(5) If f is a decreasing function of the energy alone, every difference $f_{\mathbf{p}'} - f_{\mathbf{p}}$ is positive and $P \geq 0$. Free–free gain needs either an inverted energy distribution or an anisotropic one, as for the electrons moving along the field in Eq. (5.21).

In Summary: The occupation numbers obey the master equation (6.11), derived exactly from the operator equation of motion to second order in the ion potential. No Pauli blocking appears, and the heating rate is Eq. (6.14).

7 The heating rate and the Lindhard function

7.1 The heating rate through the Lindhard function

(1) In Eq. (6.14) write the initial momentum as $\mathbf{q}$ and the final one as $\mathbf{q} + \hbar\mathbf{k}$, with $\sum_{\mathbf{q}} \to \mathcal{V}\int \mathrm{d}^3q/(2\pi\hbar)^3$ and $\sum_{\mathbf{k}} \to \mathcal{V}\int \mathrm{d}^3k/(2\pi)^3$,

$$P = \sum_{\ell=1}^{\infty} \ell\hbar\omega \, \frac{2\pi}{\hbar} \, n_i \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) \Big\{ 2 \int \frac{\mathrm{d}^3q}{(2\pi\hbar)^3} \, \delta\big(E_{\mathbf{q}+\hbar\mathbf{k}} - E_{\mathbf{q}} - \ell\hbar\omega\big) \big(f_{\mathbf{q}} - f_{\mathbf{q}+\hbar\mathbf{k}}\big) \Big\}. \tag{7.1}$$

(2) The density response function of free electrons, the Lindhard function [10], is

$$\boxed{\chi_0(\mathbf{k},\Omega) = 2 \int \frac{\mathrm{d}^3q}{(2\pi\hbar)^3} \, \frac{f_{\mathbf{q}} - f_{\mathbf{q}+\hbar\mathbf{k}}}{\hbar\Omega + E_{\mathbf{q}} - E_{\mathbf{q}+\hbar\mathbf{k}} + i0^+},} \tag{7.2}$$

the factor 2 counting the spin. With $\mathrm{Im}(x + i0^+)^{-1} = -\pi\delta(x)$ the curly bracket is $-\operatorname{Im}\chi_0(\mathbf{k},\ell\omega)/\pi$, and

$$P = -2n_i \sum_{\ell=1}^{\infty} \ell\omega \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0) \, \operatorname{Im}\chi_0(\mathbf{k},\ell\omega). \tag{7.3}$$

(3) Changing $\mathbf{q} \to \mathbf{q} + \hbar\mathbf{k}$ in Eq. (7.2) shows $\chi_0(-\mathbf{k},-\Omega)^* = \chi_0(\mathbf{k},\Omega)$, so $\operatorname{Im}\chi_0(-\mathbf{k},-\Omega) = -\operatorname{Im}\chi_0(\mathbf{k},\Omega)$. Since the rest of the integrand is even in $\mathbf{k}$, the term $-\ell$ equals the term ℓ, and the

sum can run over all ℓ,

$$\boxed{P = -n_i \sum_{\ell=-\infty}^{\infty} \ell\omega \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k})\, J_\ell^2(\mathbf{k}\cdot\mathbf{a}_0)\, \mathrm{Im}\,\chi_0(\mathbf{k},\ell\omega).} \tag{7.4}$$

This form holds for any distribution f, isotropic or not, and any field strength. Section 10 shows that the interacting electron gas gives the same formula with χ_0 replaced by its density response.

7.2 The imaginary part of the Lindhard function

(1) For an isotropic distribution $f_\mathbf{q} = f(E_\mathbf{q})$, write $\mathrm{Im}\,\chi_0 = -2\pi \int \mathrm{d}^3q\,(2\pi\hbar)^{-3}[f(E_\mathbf{q}) - f(E_{\mathbf{q}+\hbar\mathbf{k}})]\,\delta(\hbar\Omega + E_\mathbf{q} - E_{\mathbf{q}+\hbar\mathbf{k}})$. In the first term $E_{\mathbf{q}+\hbar\mathbf{k}} - E_\mathbf{q} = \hbar k q_\parallel/m + \hbar^2k^2/2m$ with $q_\parallel = \mathbf{q}\cdot\hat{\mathbf{k}}$, so the delta function is $(m/\hbar k)\,\delta(q_\parallel - mc_-)$ with $c_- = \Omega/k - \hbar k/2m$.
(2) The integral over the transverse momentum is $\int f\,\mathrm{d}^2q_\perp = \pi\int f\,\mathrm{d}(q_\perp^2) = 2\pi m\int_{E_-}^\infty f(E)\,\mathrm{d}E$, where $E_- = mc_-^2/2$ is the energy at $q_\perp = 0$.
(3) In the second term the shift $\mathbf{q} \to \mathbf{q} - \hbar\mathbf{k}$ turns the condition into $\hbar k q_\parallel/m - \hbar^2k^2/2m = \hbar\Omega$, that is $q_\parallel = mc_+$ with $c_+ = \Omega/k + \hbar k/2m$ and $E_+ = mc_+^2/2$.
(4) Subtracting the two and collecting the constants $2\pi\cdot 2\pi m\cdot m/[(2\pi\hbar)^3\hbar k] = m^2/2\pi\hbar^4 k$,

$$\boxed{\mathrm{Im}\,\chi_0(\mathbf{k},\Omega) = -\frac{m^2}{2\pi\hbar^4 k}\int_{E_-}^{E_+} f(E)\,\mathrm{d}E, \qquad E_\pm = \frac{m}{2}\Big(\frac{\Omega}{k} \pm \frac{\hbar k}{2m}\Big)^2.} \tag{7.5}$$

For $\Omega > 0$, $E_+ > E_-$ and $\mathrm{Im}\,\chi_0 < 0$.
(5) For the Fermi–Dirac distribution $f = [e^{(E-\mu)/k_BT} + 1]^{-1}$, with the antiderivative $-k_BT\ln(1 + e^{(\mu-E)/k_BT})$,

$$\boxed{\mathrm{Im}\,\chi_0(\mathbf{k},\Omega) = -\frac{m^2 k_BT}{2\pi\hbar^4 k}\,\ln\frac{1+e^{(\mu-E_-)/k_BT}}{1+e^{(\mu-E_+)/k_BT}}.} \tag{7.6}$$

(6) For nondegenerate electrons $f = e^{(\mu-E)/k_BT}$ with $e^{\mu/k_BT} = \frac{1}{2}n_e(2\pi\hbar^2/mk_BT)^{3/2}$, and the constants combine into $(\pi/2)^{1/2}/\hbar k v_t$,

$$\boxed{\mathrm{Im}\,\chi_0(\mathbf{k},\Omega) = -n_e\Big(\frac{\pi}{2}\Big)^{1/2}\frac{1}{\hbar k v_t}\Big[e^{-E_-/k_BT} - e^{-E_+/k_BT}\Big], \qquad \frac{E_\pm}{k_BT} = \frac{1}{2}\Big(\frac{\Omega}{kv_t} \pm \frac{\hbar k v_t}{2k_BT}\Big)^2.} \tag{7.7}$$

(7) In the classical limit $\hbar \to 0$ the bracket is $(\hbar\Omega/k_BT)e^{-m\Omega^2/2k^2k_BT}$, and with $\zeta = \Omega/\sqrt{2}kv_t$,

$$\mathrm{Im}\,\chi_0 \;\longrightarrow\; -\frac{n_e}{k_BT}\sqrt{\pi}\,\zeta\, e^{-\zeta^2}, \tag{7.8}$$

the Landau damping of the classical dielectric function, Eq. (11.19) [1]. The quantum form differs from it only for $\hbar k \gtrsim mv_t$, and that difference cuts off the Coulomb logarithm.

7.3 Weak fields

(1) For $|\mathbf{k}\cdot\mathbf{a}_0| \ll 1$ only $\ell = \pm 1$ contribute to Eq. (7.4), each with $J_1^2 = (\mathbf{k}\cdot\mathbf{a}_0)^2/4$,

$$P = -\frac{n_i\omega}{2}\int \frac{\mathrm{d}^3k}{(2\pi)^3}\,|\tilde{v}(\mathbf{k})|^2 S(\mathbf{k})\,(\mathbf{k}\cdot\mathbf{a}_0)^2\,\mathrm{Im}\,\chi_0(\mathbf{k},\omega). \tag{7.9}$$

(2) For an isotropic distribution and isotropic ions the rest of the integrand depends on k only, and the angular average gives $(\mathbf{k}\cdot\mathbf{a}_0)^2 \to k^2a_0^2/3$. With $\int \mathrm{d}^3k/(2\pi)^3 = (2\pi^2)^{-1}\int k^2\mathrm{d}k$,

$$P = -\frac{n_i\omega a_0^2}{12\pi^2}\int_0^\infty k^4|\tilde v(k)|^2 S(k)\ \mathrm{Im}\,\chi_0(k,\omega)\,\mathrm{d}k. \tag{7.10}$$

(3) Dividing by $n_e m\omega^2 a_0^2/2 = n_e m v_0^2/2$ and, for a Coulomb ion, using $k^4|\tilde v|^2 = 16\pi^2Z^2e^4$ and $n_iZ^2/n_e = Z$,

$$\boxed{\nu_{\mathrm{eff}}(\omega) = -\frac{n_i}{6\pi^2 n_e m\omega}\int_0^\infty k^4|\tilde v(k)|^2 S(k)\ \mathrm{Im}\,\chi_0(k,\omega)\,\mathrm{d}k = -\frac{8Ze^4}{3m\omega}\int_0^\infty S(k)\ \mathrm{Im}\,\chi_0(k,\omega)\,\mathrm{d}k.} \tag{7.11}$$

By Eq. (2.7) this collision frequency gives the transverse dielectric function $\mathrm{Im}\,\varepsilon_T = \omega_p^2\nu_{\mathrm{eff}}/\omega^3$ and the absorption coefficient of Eq. (2.9). The quantum theory has replaced the input ν of the Drude model by a formula.

8 Closed forms for weak fields

Throughout this section the ions are bare Coulomb charges, $S = 1$, and the field is weak, so that Eq. (7.11) applies.

8.1 Maxwell electrons

(1) Insert Eq. (7.7) at $\Omega = \omega$ into Eq. (7.11),

$$\nu_{\mathrm{eff}} = \frac{8Ze^4}{3m\omega}\,n_e\Big(\frac{\pi}{2}\Big)^{1/2}\frac{1}{\hbar v_t}\int_0^\infty\frac{\mathrm{d}k}{k}\Big[e^{-E_-/k_BT} - e^{-E_+/k_BT}\Big]. \tag{8.1}$$

(2) Expand the exponents of Eq. (7.7). With $mv_t^2 = k_BT$ and $u = \hbar\omega/k_BT$,

$$\frac{E_\pm}{k_BT} = \frac{\omega^2}{2k^2v_t^2} \pm \frac{u}{2} + \frac{\hbar^2k^2}{8mk_BT}, \qquad e^{-E_-/k_BT} - e^{-E_+/k_BT} = 2\sinh\frac{u}{2}\,e^{-a/k^2-bk^2}, \tag{8.2}$$

with $a = \omega^2/2v_t^2$ and $b = \hbar^2/8mk_BT$.
(3) The integral representation $K_0(z) = \frac12\int_0^\infty \exp(-t - z^2/4t)\,t^{-1}\mathrm{d}t$, Eq. (10.32.10) [11] at $\nu = 0$, with $t = bk^2$ and $\mathrm{d}t/t = 2\mathrm{d}k/k$ gives

$$\int_0^\infty\frac{\mathrm{d}k}{k}\,e^{-a/k^2-bk^2} = K_0\big(2\sqrt{ab}\big), \qquad 2\sqrt{ab} = 2\Big(\frac{\omega^2\hbar^2}{16(k_BT)^2}\Big)^{1/2} = \frac{u}{2}. \tag{8.3}$$

(4) Collect. Since $m\omega\hbar v_t = u\,m^{1/2}(k_BT)^{3/2}$, $\frac83(\pi/2)^{1/2} = \frac43(2\pi)^{1/2}$ and $Zn_e = Z^2n_i$,

$$\boxed{\nu_{\mathrm{eff}} = \nu_T\,\Lambda(u), \qquad \Lambda(u) = \frac{2\sinh(u/2)}{u}\,K_0\Big(\frac{u}{2}\Big) = \frac{1-e^{-u}}{u}\,e^{u/2}K_0\Big(\frac{u}{2}\Big), \qquad u = \frac{\hbar\omega}{k_BT},} \tag{8.4}$$

with ν_T of Eq. (2.10). The quantum theory gives the collision frequency of the Drude model with the logarithm $\ln\Lambda_e$ of Eq. (10.133) [1] replaced by the function $\Lambda(u)$, and with no cutoff.
(5) The limits follow from $K_0(x) = -\ln(x/2) - \gamma + O(x^2\ln x)$, Eq. (10.31.2) [11], and $K_0(x) \sim (\pi/2x)^{1/2}e^{-x}$, Eq. (10.40.2) [11], where $\gamma = 0.5772$ is Euler's constant,

$$\boxed{\Lambda(u) = \ln\frac{4k_BT}{e^\gamma\hbar\omega} + O(u^2\ln u) \quad (u \ll 1), \qquad \Lambda(u) \simeq \sqrt{\pi}\,u^{-3/2} \quad (u \gg 1).} \tag{8.5}$$

(6) For $u \ll 1$ the logarithm is $\ln(b_{\max}/b_{\min})$ with

$$b_{\max} = \frac{v_t}{\omega}, \qquad b_{\min} = \frac{e^\gamma}{4}\,\frac{\hbar}{mv_t} = 0.445\,\frac{\hbar}{mv_t}. \tag{8.6}$$

The upper cutoff is the distance an electron travels in a radian of the period, the dynamic cutoff of Dawson and Oberman, not the Debye length. At $\omega = \omega_p$ the two coincide, $v_t/\omega_p = \lambda_D$. The lower cutoff is fixed by the theory. The de Broglie length $\hbar/2mv$ at $mv^2 = 3k_BT$ of Eq. (10.136) [1] is $0.289\,\hbar/mv_t$, smaller by a factor 0.65, so the classical prescription overestimates the logarithm by $\ln(0.445/0.289) = 0.43$ at the critical density. Table 2 and Fig. 2 give $\Lambda(u)$ and its asymptotes.

Table 2: The quantum Coulomb logarithm of inverse bremsstrahlung for Maxwell electrons, Eq. (8.4), its two asymptotes of Eq. (8.5), and the thermally averaged Born Gaunt factor of Eq. (9.9).

u	$\Lambda(u)$	$\ln(4/e^\gamma u)$	$\sqrt{\pi}\,u^{-3/2}$	$\langle g\rangle$
0.001	7.717	7.717	56050	4.257
0.003	6.618	6.618	10787	3.654
0.01	5.414	5.414	1772	3.000
0.03	4.316	4.316	341.1	2.415
0.1	3.116	3.112	56.05	1.805
0.3	2.038	2.013	10.79	1.300
1	0.963	0.809	1.772	0.840
3	0.304	−0.290	0.341	0.528
10	0.0548	−1.494	0.0560	0.302

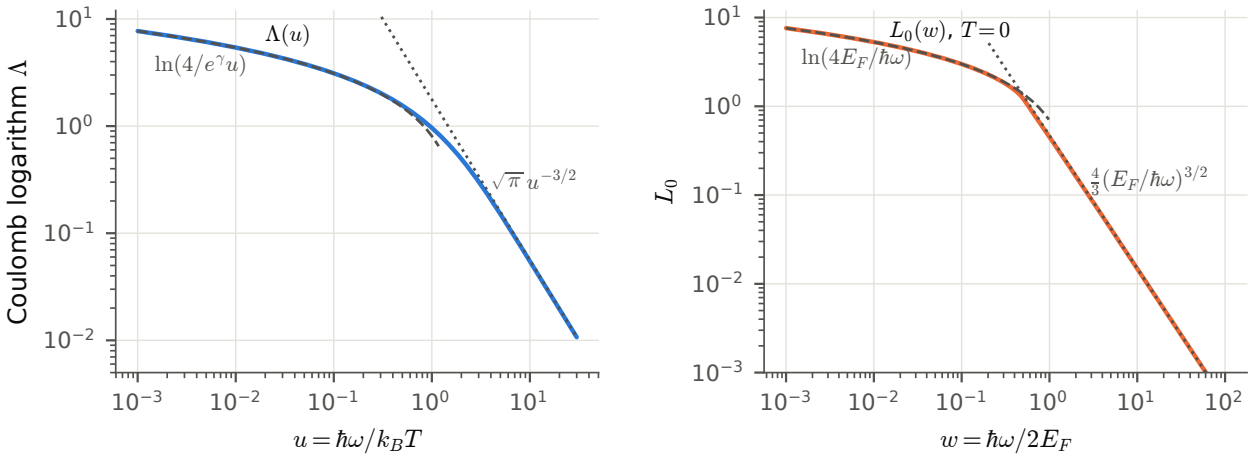


Figure 2: Left, the Coulomb logarithm $\Lambda(u)$ of Maxwell electrons, Eq. (8.4), with its asymptotes of Eq. (8.5). Right, the function $L_0(w)$ of the degenerate gas at $T = 0$, Eq. (8.11), with its asymptotes of Eq. (8.12), against $w = \hbar\omega/2E_F$.

8.2 Arbitrary degeneracy

Insert Eq. (7.6) into Eq. (7.11),

$$\boxed{\nu_{\rm eff} = \frac{4Ze^4 mk_BT}{3\pi\hbar^4\omega}\int_0^\infty \frac{\mathrm{d}k}{k}\,\ln\frac{1+e^{(\mu-E_-)/k_BT}}{1+e^{(\mu-E_+)/k_BT}}, \qquad E_\pm = \frac{m}{2}\Big(\frac{\omega}{k} \pm \frac{\hbar k}{2m}\Big)^2.} \tag{8.7}$$

For $e^{\mu/k_BT} \ll 1$ the logarithm is $e^{\mu/k_BT}(e^{-E_-/k_BT} - e^{-E_+/k_BT})$ and Eq. (8.4) follows. The chemical potential is fixed by $n_e = 2\int \mathrm{d}^3q\,(2\pi\hbar)^{-3} f$. Pauli blocking does not appear, by Section 6.4, but the occupation of the states does.

8.3 The degenerate electron gas at zero temperature

(1) At $T = 0$, $k_BT \ln[(1 + e^{(\mu-E_-)/k_BT})/(1 + e^{(\mu-E_+)/k_BT})] \to (E_F - E_-)_+ - (E_F - E_+)_+$, where $(y)_+ = \max(y, 0)$ and $\mu = E_F = mv_F^2/2$. With $x = k/k_F$, $k_F = mv_F/\hbar$ and $w = \hbar\omega/2E_F = \omega/k_Fv_F$, $E_\pm/E_F = (w/x \pm x/2)^2$ and

$$\nu_{\text{eff}} = \frac{4Ze^4mE_F}{3\pi\hbar^4\omega}\, I_0(w), \qquad I_0(w) = \int_0^\infty \frac{\mathrm{d}x}{x}\Big\{\Big[1 - \Big(\frac{w}{x} - \frac{x}{2}\Big)^2\Big]_+ - \Big[1 - \Big(\frac{w}{x} + \frac{x}{2}\Big)^2\Big]_+\Big\}. \tag{8.8}$$

(2) The first bracket is positive for $|w/x - x/2| < 1$, that is for $x_1 < x < x_2$ with $x_{1,2} = \sqrt{1+2w} \mp 1$. There the integrand is $(1+w)/x - w^2/x^3 - x/4$, with the antiderivative $(1+w)\ln x + w^2/2x^2 - x^2/8$. At the end points $w/x - x/2 = \pm 1$, and $\frac12(w^2/x^2 - x^2/4) = \frac12(w/x - x/2)(w/x + x/2)$ equals $(x_1+1)/2$ at x_1 and $-(x_2-1)/2$ at x_2. Hence

$$I_-(w) = (1+w)\ln\frac{\sqrt{1+2w}+1}{\sqrt{1+2w}-1} - \sqrt{1+2w}. \tag{8.9}$$

(3) The second bracket is positive for $w/x + x/2 < 1$, which needs $w < 1/2$ and then $x_3 < x < x_4$ with $x_{3,4} = 1 \mp \sqrt{1-2w}$. The same steps, with the integrand $(1-w)/x - w^2/x^3 - x/4$, give

$$I_+(w) = (1-w)\ln\frac{1+\sqrt{1-2w}}{1-\sqrt{1-2w}} - \sqrt{1-2w} \quad (w < \tfrac12), \qquad I_+ = 0 \quad (w \ge \tfrac12). \tag{8.10}$$

(4) With $4Ze^4mE_F/(3\pi\hbar^4\omega) = (4Ze^4m/3\pi\hbar^3)/(2w)$,

$$\boxed{\nu_{\text{eff}} = \frac{4Ze^4m}{3\pi\hbar^3}\, L_0(w), \qquad L_0(w) = \frac{I_-(w) - I_+(w)}{2w}, \qquad w = \frac{\hbar\omega}{2E_F}.} \tag{8.11}$$

(5) For $w \ll 1$, $I_-(w) - I_+(w) = 2w\ln(2/w) + O(w^2)$, and for $w \gg 1$, $I_- = \frac43(2w)^{-1/2} + O(w^{-3/2})$, so

$$\boxed{L_0(w) \simeq \ln\frac{4E_F}{\hbar\omega} \quad (\hbar\omega \ll E_F), \qquad L_0(w) \simeq \frac43\Big(\frac{E_F}{\hbar\omega}\Big)^{3/2} \quad (\hbar\omega \gg E_F).} \tag{8.12}$$

(6) The low-frequency form is the resistivity formula of Ziman. For an electron at the Fermi surface the Born cross section $(m/2\pi\hbar^2)^2|\tilde v(q)|^2$ with $q = 2k_F\sin(\theta/2)$, $1 - \cos\theta = q^2/2k_F^2$ and $\mathrm{d}\Omega = 2\pi q\,\mathrm{d}q/k_F^2$ gives the momentum relaxation rate

$$\frac{1}{\tau} = n_iv_F\int(1-\cos\theta)\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\mathrm{d}\Omega = \frac{n_im}{4\pi\hbar^3k_F^3}\int_{k_{\min}}^{2k_F} q^3|\tilde v(q)|^2\,\mathrm{d}q = \frac{4Ze^4m}{3\pi\hbar^3}\ln\frac{2k_F}{k_{\min}}, \tag{8.13}$$

where the last step uses the Coulomb form and $k_F^3 = 3\pi^2n_e = 3\pi^2Zn_i$. With the minimum momentum transfer $k_{\min} = \omega/v_F$ that energy conservation imposes on an electron at the Fermi surface, $\ln(2k_F/k_{\min}) = \ln(4E_F/\hbar\omega)$, and the two agree.

8.4 The universal high-frequency tail

(1) When $\hbar\omega$ exceeds every electron energy that is occupied, Eq. (7.5) has $E_+ = E_- + \hbar\omega$ far above the occupied energies, and E_- is small only near $k_\omega = (2m\omega/\hbar)^{1/2}$, where $\omega/k = \hbar k/2m$. There $c_- = \omega/k - \hbar k/2m \simeq -(\hbar/m)(k - k_\omega)$.
(2) Hence

$$\int_0^\infty \frac{\mathrm{d}k}{k} \int_{E_-}^{E_+} f \,\mathrm{d}E \simeq \frac{m}{\hbar k_\omega} \int_{-\infty}^{\infty} \mathrm{d}c \int_{mc^2/2}^{\infty} f(E)\,\mathrm{d}E = \frac{m}{\hbar k_\omega}\, 2 \int_0^\infty f(E) \Big(\frac{2E}{m}\Big)^{1/2} \mathrm{d}E = \frac{m}{\hbar k_\omega}\, \frac{n_e (2\pi\hbar)^3}{4\pi m^2}, \tag{8.14}$$

where the last step uses $n_e = 8\pi m (2\pi\hbar)^{-3} \int (2mE)^{1/2} f \,\mathrm{d}E$.
(3) With Eq. (7.5), $\int_0^\infty \mathrm{Im}\,\chi_0 \,\mathrm{d}k = -\pi m n_e/\hbar^2 k_\omega$ for every distribution, and Eq. (7.11) gives

$$\boxed{\nu_{\mathrm{eff}} \;\longrightarrow\; \frac{4\sqrt{2}\,\pi}{3}\, \frac{Z n_e e^4}{m^{1/2} (\hbar\omega)^{3/2}}.} \tag{8.15}$$

Both the Maxwell form, $\nu_T \sqrt{\pi} u^{-3/2}$, and the degenerate form, $(4Ze^4 m/3\pi\hbar^3)\frac{4}{3}(E_F/\hbar\omega)^{3/2}$ with $E_F^{3/2} = 3\pi^2\hbar^3 n_e/(2m)^{3/2}$, reduce to it. When the photon energy exceeds the kinetic energies the electrons absorb as if they were at rest.

Remarks:
(1) For hydrogen at $k_B T = 1$ keV and $\lambda_L = 0.351\ \mu$m, $u = 0.0035$ and $\Lambda = 6.45$, while the logarithm of the cutoffs stated with Eq. (10.136) [1] at $n_e = n_c/4$ is 7.58, and the printed formula gives 6.69. The difference between the printed formula and its stated cutoffs is taken up in Appendix A.2.
(2) Screening changes the lower limit of the k integral, where $\mathrm{Im}\,\chi_0$ is already exponentially small below $k \sim \omega/v_t$. Its effect is computed in Section 10.5 and is a few tenths in Λ near the critical density.

9 Spontaneous bremsstrahlung, the Gaunt factor and Kirchhoff's law

9.1 From stimulated to spontaneous emission

(1) Quantize one mode of the radiation field in the box, with wave vector $\mathbf{k}_L$, polarization $\hat{\mathbf{e}}_\lambda$ and frequency $\omega = ck_L$,

$$\hat{\mathbf{E}} = i\Big(\frac{2\pi\hbar\omega}{\mathcal{V}}\Big)^{1/2} \hat{\mathbf{e}}_\lambda \big(\hat{a}\, e^{i(\mathbf{k}_L\cdot\mathbf{r} - \omega t)} - \hat{a}^\dagger e^{-i(\mathbf{k}_L\cdot\mathbf{r} - \omega t)}\big). \tag{9.1}$$

In a state with N photons the absorption matrix element carries $\langle N-1|\hat{a}|N\rangle = \sqrt{N}$ and the emission one $\langle N+1|\hat{a}^\dagger|N\rangle = \sqrt{N+1}$.
(2) The classical field $E_0 \hat{\mathbf{e}} \cos\omega t$ has the amplitude $E_0/2$ in front of $e^{-i\omega t}$. For $N \gg 1$ it corresponds to $E_0/2 = (2\pi\hbar\omega N/\mathcal{V})^{1/2}$, that is $E_0^2/8\pi = N\hbar\omega/\mathcal{V}$, the energy density of N photons. The weak-field rates of Section 5.5 are therefore N times the rates for one photon, and the spontaneous emission into the mode is the stimulated rate with $N \to 1$,

$$\boxed{W^{\mathrm{sp}}_{\mathbf{p}\to\mathbf{p}'}(\mathbf{k}_L, \lambda) = W^{(\ell=-1)}_{\mathbf{p}\to\mathbf{p}'}\Big|_{E_0^2 = 8\pi\hbar\omega/\mathcal{V},\ \hat{\mathbf{e}}=\hat{\mathbf{e}}_\lambda}.} \tag{9.2}$$

This is the content of Einstein's relation, Eqs. (10.84) and (10.85) [1].

9.2 The Born bremsstrahlung cross section

(1) With $a_0^2 = e^2E_0^2/m^2\omega^4 = 8\pi\hbar e^2/(\mathcal{V}m^2\omega^3)$, the cross section of Eq. (5.13) for $\ell = -1$ and one mode is

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega'} = \frac{m^2}{4\pi^2\hbar^4}\frac{p'}{p}|\tilde{v}(\mathbf{k})|^2\,\frac{(\mathbf{k}\cdot\hat{\mathbf{e}}_\lambda)^2}{4}\,\frac{8\pi\hbar e^2}{\mathcal{V}m^2\omega^3} = \frac{e^2}{2\pi\hbar^3\mathcal{V}\omega^3}\,\frac{p'}{p}\,|\tilde{v}(\mathbf{k})|^2(\mathbf{k}\cdot\hat{\mathbf{e}}_\lambda)^2, \tag{9.3}$$

with $p' = (p^2 - 2m\hbar\omega)^{1/2}$.
(2) Sum over the modes in vacuum, $\sum_{\mathbf{k}_L,\lambda} \to \mathcal{V}\sum_\lambda \int \omega^2\mathrm{d}\omega\,\mathrm{d}\Omega_L/(2\pi c)^3$. The two polarizations perpendicular to $\hat{\mathbf{k}}_L$ give $\sum_\lambda(\mathbf{k}\cdot\hat{\mathbf{e}}_\lambda)^2 = k^2 - (\mathbf{k}\cdot\hat{\mathbf{k}}_L)^2$, whose integral over the directions of the photon is $4\pi k^2 - \frac{4\pi}{3}k^2 = \frac{8\pi}{3}k^2$. The emission cross section per unit frequency is

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\omega\,\mathrm{d}\Omega'} = \frac{e^2}{2\pi\hbar^3\omega^3}\,\frac{\omega^2}{8\pi^3c^3}\,\frac{8\pi}{3}\,\frac{p'}{p}\,k^2|\tilde{v}(\mathbf{k})|^2 = \frac{e^2}{6\pi^3\hbar^3c^3\omega}\,\frac{p'}{p}\,k^2|\tilde{v}(\mathbf{k})|^2. \tag{9.4}$$

(3) For a Coulomb ion $k^2|\tilde{v}|^2 = 16\pi^2Z^2e^4/k^2$, and $\int \mathrm{d}\Omega'/k^2 = 2\pi\hbar^2\int_{-1}^{1}\mathrm{d}c/(p^2 + p'^2 - 2pp'c) = (2\pi\hbar^2/pp')L$ with $L = \ln[(p+p')/(p-p')]$, so

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\omega} = \frac{16}{3}\,\frac{Z^2e^6}{m^2c^3\hbar v^2\omega}\,\ln\frac{p+p'}{p-p'}.} \tag{9.5}$$

This is the nonrelativistic Born cross section of bremsstrahlung, $\mathrm{d}\sigma = \frac{16}{3}Z^2\alpha r_e^2(c^2/v^2)L\,\mathrm{d}\omega/\omega$ with $\alpha = e^2/\hbar c$ and $r_e = e^2/mc^2$.
(4) Kramers' cross section of Eq. (2.11), per unit angular frequency, is $(\mathrm{d}\sigma/\mathrm{d}\nu)/2\pi$ with $h\nu = \hbar\omega$, that is $(16\pi/3\sqrt{3})Z^2e^6/(m^2c^3v^2\hbar\omega)$. The ratio is the Born Gaunt factor,

$$\boxed{g_B(E,\hbar\omega) = \frac{\sqrt{3}}{\pi}\ln\frac{p+p'}{p-p'}, \qquad p = (2mE)^{1/2}, \quad p' = [2m(E-\hbar\omega)]^{1/2}.} \tag{9.6}$$

For soft photons $g_B \to (\sqrt{3}/\pi)\ln(4E/\hbar\omega)$, and at the threshold $p' \to 0$ it vanishes as $(\sqrt{3}/\pi)2p'/p$.

9.3 The thermal average

(1) The emission of a Maxwellian plasma is proportional to $\int \mathrm{d}^3v\,f\,v\,\mathrm{d}\sigma/\mathrm{d}\omega \propto \int v\,\mathrm{d}v\,e^{-E/k_BT}g_B = m^{-1}\int \mathrm{d}E\,e^{-E/k_BT}g_B$ over $E > \hbar\omega$, while Kramers' form has $g = 1$. The thermally averaged Gaunt factor is therefore

$$\langle g\rangle = \frac{\int_{\hbar\omega}^{\infty} e^{-E/k_BT}g_B\,\mathrm{d}E}{\int_{\hbar\omega}^{\infty} e^{-E/k_BT}\mathrm{d}E} = \frac{\sqrt{3}}{\pi}\,e^u\int_u^\infty e^{-\epsilon}\ln\frac{\sqrt{\epsilon}+\sqrt{\epsilon-u}}{\sqrt{\epsilon}-\sqrt{\epsilon-u}}\,\mathrm{d}\epsilon, \tag{9.7}$$

with $\epsilon = E/k_BT$.
(2) Substitute $\epsilon = u\cosh^2 t$. Then $\sqrt{\epsilon}\pm\sqrt{\epsilon-u} = \sqrt{u}\,e^{\pm t}$, the logarithm is $2t$, and $\mathrm{d}\epsilon = u\sinh 2t\,\mathrm{d}t$. With $\cosh^2 t = \frac{1}{2}(1+\cosh 2t)$ and $\tau = 2t$,

$$\int_u^\infty e^{-\epsilon}\ln(\cdots)\,\mathrm{d}\epsilon = \frac{u}{2}\,e^{-u/2}\int_0^\infty \tau\sinh\tau\,e^{-(u/2)\cosh\tau}\mathrm{d}\tau. \tag{9.8}$$

(3) Integrate by parts with $\sinh\tau\,e^{-(u/2)\cosh\tau} = \mathrm{d}[-(2/u)e^{-(u/2)\cosh\tau}]/\mathrm{d}\tau$. The boundary terms vanish, and $\int_0^\infty e^{-z\cosh\tau}\mathrm{d}\tau = K_0(z)$, Eq. (10.32.9) [11], leaves $(2/u)K_0(u/2)$. Hence

$$\boxed{\langle g\rangle(u) = \frac{\sqrt{3}}{\pi}\,e^{u/2}K_0\Big(\frac{u}{2}\Big).} \tag{9.9}$$

(4) The power radiated at all frequencies weights the spectrum with e^{-u}, the factor of Kramers' form, so its Gaunt factor is

$$\boxed{\bar{g} = \int_0^\infty \langle g\rangle(u)\, e^{-u}\, \mathrm{d}u = \frac{\sqrt{3}}{\pi}\, 2 \int_0^\infty e^{-x} K_0(x)\, \mathrm{d}x = \frac{2\sqrt{3}}{\pi} = 1.103,} \tag{9.10}$$

using $K_0(x) = \int_0^\infty e^{-x\cosh t}\mathrm{d}t$, so that $\int_0^\infty e^{-x}K_0(x)\mathrm{d}x = \int_0^\infty \mathrm{d}t/(1+\cosh t) = [\tanh(t/2)]_0^\infty = 1$. This is the factor $g \approx 2\sqrt{3}/\pi$ that Atzeni and Meyer-ter-Vehn apply to Eq. (10.93) [1]. It is the Born value exactly.

9.4 Kirchhoff's law and the Coulomb logarithm as a Gaunt factor

(1) Compare Eq. (9.9) with Eq. (8.4),

$$\boxed{\Lambda(u) = \frac{\pi}{\sqrt{3}}\, \langle g\rangle(u)\, \frac{1 - e^{-u}}{u}.} \tag{9.11}$$

The quantum Coulomb logarithm of inverse bremsstrahlung is $\pi/\sqrt{3}$ times the thermally averaged Gaunt factor, times the stimulated-emission factor divided by u.
(2) The emission coefficient per unit volume, frequency and solid angle follows from Eq. (9.5) and the steps of the thermal average,

$$\begin{aligned} j_\omega &= \frac{n_e n_i \hbar\omega}{4\pi} \left\langle v\, \frac{\mathrm{d}\sigma}{\mathrm{d}\omega} \right\rangle, \qquad \left\langle \frac{L}{v} \right\rangle = \left(\frac{2}{\pi}\right)^{1/2} \frac{\pi}{\sqrt{3}}\, \frac{\langle g\rangle e^{-u}}{v_t}, \\ j_\omega &= \frac{4}{3\pi}\left(\frac{2}{\pi}\right)^{1/2} \frac{n_e n_i Z^2 e^6}{m^2 c^3 v_t}\, \frac{\pi}{\sqrt{3}} \langle g\rangle e^{-u}. \end{aligned} \tag{9.12}$$

The absorption coefficient of Eq. (2.9) with $n_r = 1$ and $\nu_{\rm eff} = \nu_T \Lambda$, multiplied by the Planck function per unit angular frequency and solid angle, $B_\omega = (\hbar\omega^3/4\pi^3c^2)/(e^u - 1)$, is, by Eq. (9.11),

$$K\, B_\omega = \frac{\nu_T \omega_p^2}{c\omega^2}\, \frac{\pi}{\sqrt{3}} \langle g\rangle \frac{e^{-u}}{u}\, \frac{\hbar\omega^3}{4\pi^3 c^2} = \frac{4\sqrt{2\pi}}{3\pi^2}\, \frac{n_e n_i Z^2 e^6}{m^{3/2}(k_B T)^{1/2} c^3}\, \frac{\pi}{\sqrt{3}} \langle g\rangle e^{-u}. \tag{9.13}$$

Since $m^2 v_t = m^{3/2}(k_B T)^{1/2}$ and $\frac{4}{3\pi}(2/\pi)^{1/2} = \frac{4\sqrt{2\pi}}{3\pi^2}$,

$$\boxed{j_\omega = K\, B_\omega.} \tag{9.14}$$

Kirchhoff's law holds exactly between the absorption of the master equation and the spontaneous emission of Eq. (9.5), because both come from the same matrix element and the Maxwellian satisfies detailed balance.

In Summary: The free–free emission and the inverse bremsstrahlung absorption of the quantum theory are Kramers' expressions multiplied by the Born Gaunt factor. Its thermal average is $(\sqrt{3}/\pi)e^{u/2}K_0(u/2)$, its power average is $2\sqrt{3}/\pi$, and the Coulomb logarithm of the absorption is the Gaunt factor in disguise, Eq. (9.11).

10 The interacting electron gas and dynamic screening

10.1 The oscillating frame for many electrons

(1) The Hamiltonian of N electrons in the laser field and the field of the ions is

$$H = \sum_{a=1}^{N} \frac{1}{2m}\Big[\hat{\mathbf{p}}_a + \frac{e}{c}\mathbf{A}(t)\Big]^2 + \sum_{a<b} \frac{e^2}{|\hat{\mathbf{r}}_a - \hat{\mathbf{r}}_b|} + \sum_a V(\hat{\mathbf{r}}_a), \tag{10.1}$$

apart from the constant energy of the fixed ions.
(2) Let $\hat{T} = \exp[-(i/\hbar)\boldsymbol{\alpha}(t)\cdot\hat{\mathbf{P}}]$ with $\hat{\mathbf{P}} = \sum_a \hat{\mathbf{p}}_a$. It translates all electrons together, so the electron–electron interaction, which depends on differences, is unchanged, $\hat{T}^\dagger V_{ee}\hat{T} = V_{ee}$. The steps of Section 3.3, with $\dot{S} = Ne^2A^2/2mc^2$, give

$$\boxed{H' = \sum_a \frac{\hat{p}_a^2}{2m} + \sum_{a<b} \frac{e^2}{|\hat{\mathbf{r}}_a - \hat{\mathbf{r}}_b|} + \sum_a V\big(\hat{\mathbf{r}}_a + \boldsymbol{\alpha}(t)\big).} \tag{10.2}$$

In the oscillating frame the electron gas carries no field, and the ions are external charges that move along $\mathbf{R}_j - \boldsymbol{\alpha}(t)$. A uniform field couples only to the center of mass of the electrons [14], which is the content of this transformation.

10.2 The absorbed power in the two frames

(1) The power that the laser delivers to the electrons in the laboratory is $P_{\rm lab}(t) = -e\mathbf{E}(t)\cdot\sum_a\langle\mathbf{v}_a\rangle$, with the velocity $\mathbf{v}_a = (\hat{\mathbf{p}}_a + e\mathbf{A}/c)/m$. Since $\hat{T}$ commutes with $\hat{\mathbf{p}}_a$, $\sum_a\langle\mathbf{v}_a\rangle = \langle\hat{\mathbf{P}}\rangle'/m + N\dot{\boldsymbol{\alpha}}$, where the prime marks an average in the oscillating frame. The term $-eN\mathbf{E}\cdot\dot{\boldsymbol{\alpha}} \propto \cos\omega t\sin\omega t$ averages to zero over a period.
(2) In the oscillating frame, $\mathrm{d}\langle H'\rangle/\mathrm{d}t = \langle\partial_t H'\rangle = -\dot{\boldsymbol{\alpha}}\cdot\langle\mathbf{F}\rangle'$, with the force $\mathbf{F} = -\sum_a \nabla V(\hat{\mathbf{r}}_a + \boldsymbol{\alpha})$ of the ions on the electrons. The total momentum obeys $\mathrm{d}\langle\hat{\mathbf{P}}\rangle'/\mathrm{d}t = \langle\mathbf{F}\rangle'$, because the kinetic term commutes with $\hat{\mathbf{P}}$ and the internal forces cancel. With $\ddot{\boldsymbol{\alpha}} = -e\mathbf{E}/m$,

$$\frac{\mathrm{d}\langle H'\rangle}{\mathrm{d}t} = -\dot{\boldsymbol{\alpha}}\cdot\frac{\mathrm{d}\langle\hat{\mathbf{P}}\rangle'}{\mathrm{d}t} = -\frac{\mathrm{d}}{\mathrm{d}t}\big(\dot{\boldsymbol{\alpha}}\cdot\langle\hat{\mathbf{P}}\rangle'\big) - \frac{e}{m}\mathbf{E}\cdot\langle\hat{\mathbf{P}}\rangle'. \tag{10.3}$$

(3) In the steady state $\langle\hat{\mathbf{P}}\rangle'$ is periodic, and the total derivative averages to zero. Hence

$$\boxed{\langle P_{\rm lab}\rangle_t = \Big\langle\frac{\mathrm{d}\langle H'\rangle}{\mathrm{d}t}\Big\rangle_t.} \tag{10.4}$$

The absorbed power is the rate at which the energy of the electron gas grows in the oscillating frame.

10.3 Linear response to the oscillating ions

(1) With the density operator $\hat{n}_{\mathbf{k}} = \sum_a e^{-i\mathbf{k}\cdot\hat{\mathbf{r}}_a}$ and Eq. (3.8), the ion term of Eq. (10.2) is

$$\sum_a V(\hat{\mathbf{r}}_a + \boldsymbol{\alpha}) = \frac{1}{\mathcal{V}}\sum_{\mathbf{k}} U_{\mathbf{k}}(t)\,\hat{n}_{-\mathbf{k}}, \qquad U_{\mathbf{k}}(t) = \tilde{V}(\mathbf{k})\,e^{i\mathbf{k}\cdot\boldsymbol{\alpha}(t)} = \tilde{V}(\mathbf{k})\sum_\ell i^\ell J_\ell(\mathbf{k}\cdot\mathbf{a}_0)\,e^{i\ell\omega t}, \tag{10.5}$$

by Eq. (4.7). In this frame the ion potential is a sum of harmonics $\ell\omega$ with the weights J_ℓ.

(2) Let the electron gas start in equilibrium. To first order in U, Kubo's formula gives $\delta\langle\hat n_{\mathbf k}\rangle'(t) = \int \chi(\mathbf k, t-t')U_{\mathbf k}(t')\mathrm dt'$ with the retarded density response

$$\chi(\mathbf k,t) = -\frac{i}{\hbar}\,\theta(t)\,\frac{1}{\mathcal V}\big\langle[\hat n_{\mathbf k}(t),\hat n_{-\mathbf k}(0)]\big\rangle, \qquad \chi(\mathbf k,\Omega) = \int e^{i\Omega t}\chi(\mathbf k,t)\,\mathrm dt, \tag{10.6}$$

which for free electrons is the Lindhard function of Eq. (7.2) [15]. A harmonic $e^{i\ell\omega t} = e^{-i(-\ell\omega)t}$ is answered with the factor $\chi(\mathbf k, -\ell\omega)$.
(3) The power in the oscillating frame is $\mathrm d\langle H'\rangle/\mathrm dt = \mathcal V^{-1}\sum_{\mathbf k}\partial_t U_{\mathbf k}\,\langle\hat n_{-\mathbf k}\rangle'$. The term $\mathbf k = 0$ is constant. With $U_{-\mathbf k} = \tilde V(-\mathbf k)\sum_\ell (-i)^\ell J_\ell(\mathbf k\cdot\mathbf a_0)e^{i\ell\omega t}$, the response $\delta\langle\hat n_{-\mathbf k}\rangle' = \sum_\ell \chi(-\mathbf k,-\ell\omega)U_{-\mathbf k,\ell}e^{i\ell\omega t}$, and $\partial_t U_{\mathbf k} = \sum_{\ell'} i\ell'\omega U_{\mathbf k,\ell'}e^{i\ell'\omega t}$, the period average keeps $\ell' = -\ell$. The product of the coefficients is $\tilde V(\mathbf k)\tilde V(-\mathbf k)\,i^{-\ell}J_{-\ell}(-i)^\ell J_\ell = |\tilde V(\mathbf k)|^2 J_\ell^2$, and with $\chi(-\mathbf k,-\Omega) = \chi(\mathbf k,\Omega)^*$,

$$\Big\langle\frac{\mathrm d\langle H'\rangle}{\mathrm dt}\Big\rangle_t = \frac{1}{\mathcal V}\sum_{\mathbf k}\sum_\ell(-i\ell\omega)|\tilde V(\mathbf k)|^2J_\ell^2\,\chi(\mathbf k,\ell\omega)^* = -\frac{1}{\mathcal V}\sum_{\mathbf k}\sum_\ell \ell\omega\,|\tilde V(\mathbf k)|^2J_\ell^2\,\mathrm{Im}\,\chi(\mathbf k,\ell\omega), \tag{10.7}$$

where the terms with $\mathrm{Re}\,\chi$, which is even in the frequency for the isotropic gas, cancel between ℓ and $-\ell$.
(4) After the average over the ion positions and per unit volume, by Eq. (10.4),

$$\boxed{P = -n_i\sum_{\ell=-\infty}^{\infty}\ell\omega\int\frac{\mathrm d^3k}{(2\pi)^3}\,|\tilde v(\mathbf k)|^2S(\mathbf k)\,J_\ell^2(\mathbf k\cdot\mathbf a_0)\,\mathrm{Im}\,\chi(\mathbf k,\ell\omega).} \tag{10.8}$$

This is Eq. (7.4) with χ_0 replaced by the density response of the interacting gas. For free electrons the two derivations agree term by term, because the Volkov states of the laboratory are the plane waves of the oscillating frame. The result is exact to second order in the ion potential and exact in the field.

10.4 The dielectric form

(1) The longitudinal dielectric function is defined by $1/\varepsilon(\mathbf k,\Omega) = 1 + v_c(k)\chi(\mathbf k,\Omega)$ with $v_c = 4\pi e^2/k^2$. In the random-phase approximation $\varepsilon = 1 - v_c\chi_0$ and $\chi = \chi_0/\varepsilon$. For a Coulomb ion, $|\tilde v|^2\,\mathrm{Im}\,\chi = (4\pi Ze^2/k^2)^2(k^2/4\pi e^2)\,\mathrm{Im}\,\varepsilon^{-1} = (4\pi Z^2e^2/k^2)\,\mathrm{Im}\,\varepsilon^{-1}$, and

$$\boxed{P = -n_iZ^2e^2\sum_{\ell=-\infty}^{\infty}\ell\omega\int\frac{\mathrm d^3k}{(2\pi)^3}\,\frac{4\pi}{k^2}\,S(\mathbf k)\,J_\ell^2(\mathbf k\cdot\mathbf a_0)\,\mathrm{Im}\,\frac{1}{\varepsilon(\mathbf k,\ell\omega)}.} \tag{10.9}$$

With the classical dielectric function this is Silin's result [9]. With the Lindhard dielectric function it is the quantum form obtained by Kull and Plagne [16] and by Bornath and co-workers [17] from the quantum Vlasov and the quantum kinetic equation.
(2) For weak fields Eq. (7.11) with $k^4|\tilde v|^2\,\mathrm{Im}\,\chi = 4\pi Z^2e^2k^2\,\mathrm{Im}\,\varepsilon^{-1}$ gives

$$\boxed{\nu_{\mathrm{eff}}(\omega) = -\frac{2Ze^2}{3\pi m\omega}\int_0^\infty k^2S(k)\,\mathrm{Im}\,\frac{1}{\varepsilon(k,\omega)}\,\mathrm dk,} \tag{10.10}$$

the high-frequency collision frequency of Dawson and Oberman [6].

10.5 Dynamic screening for Maxwell electrons

(1) For nondegenerate electrons the Lindhard function follows from Eq. (7.2) with the shift $\mathbf{q} \to \mathbf{q} - \hbar\mathbf{k}$ in its second term and the integral over $\mathbf{q}_\perp$, which leaves the one-dimensional Maxwellian $g_1(v) = (2\pi v_t^2)^{-1/2} e^{-v^2/2v_t^2}$,

$$\chi_0 = \frac{n_e}{\hbar k} \int g_1(v) \Big[\frac{1}{c_- - v + i0^+} - \frac{1}{c_+ - v + i0^+} \Big] \mathrm{d}v = \frac{n_e}{\sqrt{2}\,\hbar k v_t} \big[Z(\zeta_+) - Z(\zeta_-) \big], \qquad \zeta_\pm = \frac{c_\pm}{\sqrt{2} v_t}, \tag{10.11}$$

with $c_\pm = \Omega/k \pm \hbar k/2m$ and the plasma dispersion function $Z(\zeta) = \pi^{-1/2} \int e^{-t^2} (t - \zeta)^{-1} \mathrm{d}t$ for $\mathrm{Im}\,\zeta > 0$ [19]. Its imaginary part on the real axis is $\sqrt{\pi} e^{-\zeta^2}$, which gives back Eq. (7.7).
(2) In the classical limit $Z(\zeta_+) - Z(\zeta_-) \to (\hbar k/\sqrt{2} m v_t) Z'(\zeta)$ with $Z' = -2(1 + \zeta Z)$, so $\chi_0 \to -(n_e/k_B T)[1 + \zeta Z(\zeta)]$ and $\varepsilon = 1 + (k_D^2/k^2)[1 + \zeta Z(\zeta)]$ with $k_D^2 = 4\pi e^2 n_e / k_B T$. This is Eq. (11.19) [1], whose function W is $W(x) = 1 + (x/\sqrt{2}) Z(x/\sqrt{2})$.
(3) With $x = k v_t/\omega$, $r = \omega_p/\omega$, $\hbar k v_t / k_B T = ux$ and $k_D^2/k^2 = r^2/x^2$,

$$\varepsilon(x) = 1 - \frac{r^2}{\sqrt{2}\,ux^3} \big[Z(\zeta_+) - Z(\zeta_-) \big], \qquad \zeta_\pm = \frac{1}{\sqrt{2}} \Big(\frac{1}{x} \pm \frac{ux}{2} \Big). \tag{10.12}$$

Equation (10.10) with $S = 1$, divided by ν_T, becomes

$$\boxed{\nu_{\mathrm{eff}} = \nu_T \Lambda_{\mathrm{RPA}}(r, u), \qquad \Lambda_{\mathrm{RPA}}(r, u) = -\frac{1}{r^2} \Big(\frac{2}{\pi} \Big)^{1/2} \int_0^\infty x^2 \ \mathrm{Im} \frac{1}{\varepsilon(x)} \, \mathrm{d}x.} \tag{10.13}$$

Without screening $\mathrm{Im}\,\varepsilon^{-1} \to -\mathrm{Im}\,\varepsilon$, the integral becomes $u^{-1} \int x^{-1} [e^{-\zeta_-^2} - e^{-\zeta_+^2}] \mathrm{d}x$, and Eq. (8.4) returns.
(4) Table 3 and Fig. 3 give the screening correction. It is negative and nearly independent of u. It is -0.10 at $n_e = n_c/4$ and -0.47 at $n_e = 0.94\, n_c$, and it approaches -0.56 as $\omega \to \omega_p$ for $u \ll 1$, slowly, because the resonance described next carries a weight proportional to $(1 - r^2)^{1/2}$. For $\omega > \omega_p$ the function $\mathrm{Re}\,\varepsilon$ vanishes at the Langmuir wave number, approximately $x_{\mathrm{pl}}^2 = (r^{-2} - 1)/3$, where the ion oscillation excites plasma waves. That resonance is part of the integral. Its weight, estimated with the Bohm–Gross form $\mathrm{Re}\,\varepsilon \simeq 1 - r^2(1 + 3x^2)$, is $\sqrt{2\pi}\, x_{\mathrm{pl}}/6r^4$, of order 0.1 for $r \geq 0.9$. For smaller r the plasma waves at x_{pl} are strongly Landau damped and there is no separate resonance.

Table 3: The screening correction $\Lambda_{\mathrm{RPA}} - \Lambda$ of the Coulomb logarithm, Eqs. (10.13) and (8.4), against $r = \omega_p/\omega = (n_e/n_c)^{1/2}$.

u	$r = 0.95$	0.9	0.8	0.7	0.5	0.3	0.1
0.003	−0.440	−0.375	−0.278	−0.204	−0.099	−0.034	−0.004
0.03	−0.439	−0.374	−0.277	−0.203	−0.098	−0.034	−0.004
0.3	−0.393	−0.333	−0.244	−0.178	−0.085	−0.029	−0.003

Remarks:
(1) Without ions H' does not depend on time and nothing is absorbed. Collisions between electrons alone do not absorb light in the dipole approximation. They enter only through χ, by screening, and through the shape of f.
(2) Equation (10.8) holds for any response function. Local-field corrections and strong coupling enter through χ, and correlated ions through $S(k)$.

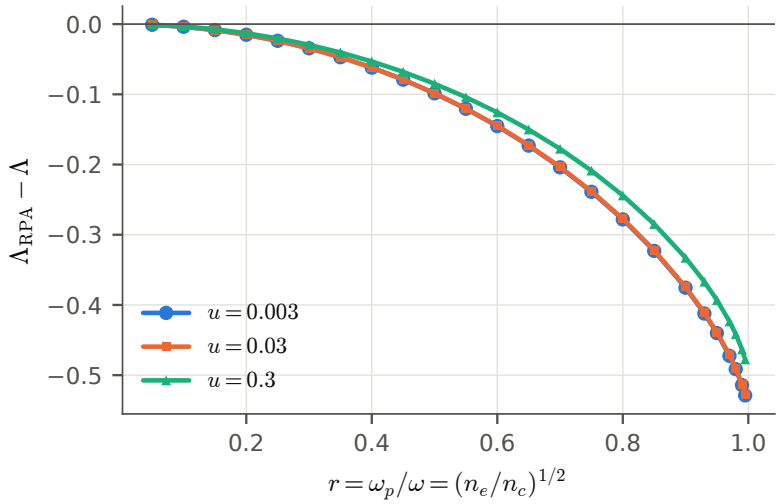


Figure 3: The screening correction $\Lambda_{\mathrm{RPA}} - \Lambda$ of Table 3 against $r = \omega_p/\omega$ for three photon energies.

(3) The same result follows from the Kadanoff–Baym equations with the dynamically screened Born self-energy [18, 17], which is the many-body counterpart of the nonequilibrium Green-function route of Section 3 of the quantum Boltzmann equation paper [4].

11 Strong fields

11.1 The exact heating rate for Maxwell electrons

(1) Take Eq. (7.4) with bare Coulomb ions, $S = 1$ and an isotropic distribution. At fixed k the direction of $\mathbf{k}$ enters only through $\mu = \hat{\mathbf{k}} \cdot \hat{\mathbf{e}}$, and $\int \mathrm{d}\Omega_k J_\ell^2(ka_0\mu) = 2\pi B_\ell(ka_0)$ with

$$B_\ell(z) = \int_{-1}^{1} J_\ell^2(z\mu)\,\mathrm{d}\mu = \frac{2}{z}\int_0^z J_\ell^2(t)\,\mathrm{d}t. \tag{11.1}$$

With $|\tilde{v}|^2 = 16\pi^2 Z^2 e^4/k^4$,

$$P = -4n_i Z^2 e^4 \sum_\ell \ell\omega \int_0^\infty \frac{\mathrm{d}k}{k^2}\,\mathrm{Im}\,\chi_0(k, \ell\omega)\,B_\ell(ka_0). \tag{11.2}$$

(2) With $x = kv_t/\omega$ and $\beta = v_0/v_t$, $ka_0 = x\beta$, and Eq. (7.7) at $\Omega = \ell\omega$ has $E_\pm/k_BT = \frac{1}{2}(\ell/x \pm ux/2)^2$ and $\hbar k v_t = ux\,k_BT$. Dividing by $n_e m v_0^2/2$ and by ν_T,

$$\boxed{\begin{aligned} &\nu_{\mathrm{eff}} = \nu_T\,\frac{3\,\mathcal{S}(\beta,u)}{u\beta^2}, \qquad \mathcal{S}(\beta,u) = \sum_{\ell=-\infty}^{\infty} \ell \int_0^\infty \frac{\mathrm{d}x}{x^3}\,D_\ell(x)\,B_\ell(x\beta), \\ &D_\ell(x) = e^{-\frac{1}{2}(\frac{\ell}{x} - \frac{ux}{2})^2} - e^{-\frac{1}{2}(\frac{\ell}{x} + \frac{ux}{2})^2}. \end{aligned}} \tag{11.3}$$

The terms $\pm\ell$ are equal. For $\beta \to 0$, $B_{\pm 1} \to (x\beta)^2/6$, and Eq. (8.3) gives the weak-field limit on the left below, which is Eq. (8.4). The nonlinearity is measured by the ratio on the right,

$$\mathcal{S} \xrightarrow{\beta\to 0} \frac{\beta^2}{3}\,2\sinh\frac{u}{2}\,K_0\Big(\frac{u}{2}\Big), \qquad R(\beta,u) = \frac{\nu_{\mathrm{eff}}(v_0)}{\nu_{\mathrm{eff}}(0)} = \frac{3\,\mathcal{S}(\beta,u)}{u\beta^2\Lambda(u)}. \tag{11.4}$$

(3) The functions B_ℓ follow from a recursion. The recurrences of the Bessel functions, Eq. (10.6.2) [11], give $\mathrm{d}[z(J_\ell^2 + J_{\ell+1}^2)]/\mathrm{d}z = (2\ell+1)(J_\ell^2 - J_{\ell+1}^2)$, so

$$\int_0^z J_\ell^2 \, \mathrm{d}t = \int_0^z J_{\ell+1}^2 \, \mathrm{d}t + \frac{z\,[J_\ell^2(z) + J_{\ell+1}^2(z)]}{2\ell+1}, \tag{11.5}$$

which is summed downward from an order $\ell \gg z$ at which the integral vanishes. The terms added are positive, and the recursion is stable.

11.2 The quasi-static limit

(1) For $x \gg 1$, that is $k \gg \omega/v_t$, the function $\ell \mapsto \ell D_\ell(x)$ changes on the scale $\Delta\ell \sim x$, much larger than one, and the limit of Eq. (5.23) applies to the sum over ℓ at fixed k,

$$\sum_\ell \ell \, D_\ell(x) \, J_\ell^2(x\beta\mu) \;\longrightarrow\; \frac{1}{2\pi}\int_0^{2\pi} \nu \, D_\nu(x) \, \mathrm{d}\theta, \qquad \nu = x\beta\mu\cos\theta. \tag{11.6}$$

(2) In dimensional form the frequency $\ell\omega$ becomes the instantaneous Doppler frequency $\Omega_t = \mathbf{k}\cdot\mathbf{v}_{\mathrm{osc}}(t)$, up to a sign that does not matter because $\Omega \,\mathrm{Im}\,\chi_0(k,\Omega)$ is even in Ω,

$$\boxed{P \;\longrightarrow\; -n_i \int \frac{\mathrm{d}^3k}{(2\pi)^3} \, |\tilde{v}(\mathbf{k})|^2 S(\mathbf{k}) \, \big\langle \Omega_t \,\mathrm{Im}\,\chi_0(\mathbf{k},\Omega_t)\big\rangle_t, \qquad \Omega_t = \mathbf{k}\cdot\mathbf{v}_{\mathrm{osc}}(t).} \tag{11.7}$$

The ions move through the electron gas of the oscillating frame with the velocity $-\mathbf{v}_{\mathrm{osc}}(t)$, and Eq. (11.7) is the stopping power of the dielectric theory of Section 11.5 of [1] at that velocity, times the velocity, averaged over the period. This is Silin's picture.

(3) The limit fails for $k \lesssim \omega/v_t$. There every term of the exact sum carries $e^{-\ell^2/2x^2} \le e^{-1/2x^2}$ and is exponentially small, even when $v_0 \gg v_t$, because the electron gas of the oscillating frame is thermal and the ions supply only the frequencies $\ell\omega$. The exact sum therefore keeps the lower cutoff $k \sim \omega/v_t$ at all field strengths, while the quasi-static formula needs it put in.

11.3 The classical evaluation and Silin's formula

(1) In the classical limit, Eq. (7.8) with $\zeta = \Omega_t/\sqrt{2}kv_t = s\mu$, where $s = s_0\cos\theta$ and $s_0 = v_0/\sqrt{2}v_t$, gives $\Omega_t \,\mathrm{Im}\,\chi_0 = -(n_e/k_BT)\sqrt{\pi}\,\sqrt{2}kv_t\,\zeta^2 e^{-\zeta^2}$. The integrand of Eq. (11.7) is then k^{-3} times a function of direction, and the logarithmic integral $\int \mathrm{d}k/k$ between the two cutoffs is the Coulomb logarithm, held fixed here.

(2) The angular integral is

$$\int \mathrm{d}\Omega_k \, \zeta^2 e^{-\zeta^2} = 2\pi \int_{-1}^{1} s^2\mu^2 e^{-s^2\mu^2} \mathrm{d}\mu = \pi g(s), \qquad g(s) = 4s^2 \int_0^1 \tau^2 e^{-s^2\tau^2} \mathrm{d}\tau = \frac{\sqrt{\pi}\,\mathrm{erf}\,s}{s} - 2e^{-s^2}, \tag{11.8}$$

where the last form follows by one integration by parts. For small s, $g = \frac{4}{3}s^2 - \frac{4}{5}s^4 + \frac{2}{7}s^6 + O(s^8)$.

(3) At fixed logarithm the heating rate is proportional to $\langle g(s_0\cos\theta)\rangle_\theta$, and its weak-field value is $\frac{4}{3}s_0^2\langle\cos^2\theta\rangle = \frac{2}{3}s_0^2$. The ratio of the strong-field collision frequency to the weak-field one is therefore

$$\boxed{G(s_0) = \frac{3\,\langle g(s_0\cos\theta)\rangle_\theta}{2s_0^2}, \qquad s_0 = \frac{v_0}{\sqrt{2}\,v_t}.} \tag{11.9}$$

(4) Silin's form follows from the integral representation of g. With $\langle e^{-a\cos^2\theta}\rangle = e^{-a/2}I_0(a/2)$, from Eq. (10.32.1) [11], and $I_0' = I_1$, Eq. (10.29.3) [11],

$$\big\langle \cos^2\theta\, e^{-a\cos^2\theta}\big\rangle = -\frac{\mathrm{d}}{\mathrm{d}a}\Big[e^{-a/2}I_0\Big(\frac{a}{2}\Big)\Big] = \frac{1}{2}e^{-a/2}\Big[I_0\Big(\frac{a}{2}\Big) - I_1\Big(\frac{a}{2}\Big)\Big]. \tag{11.10}$$

With $a = s_0^2\tau^2$ and $z = s_0\tau/\sqrt{2}$,

$$\begin{aligned} \langle g(s_0\cos\theta)\rangle &= 4s_0^2\int_0^1 \tau^2\big\langle \cos^2\theta\, e^{-s_0^2\tau^2\cos^2\theta}\big\rangle \mathrm{d}\tau = \frac{4\sqrt{2}}{s_0}\,Q(r), \\ Q(r) &= \int_0^r z^2 e^{-z^2}\big[I_0(z^2) - I_1(z^2)\big]\mathrm{d}z, \end{aligned} \tag{11.11}$$

with $r = s_0/\sqrt{2} = v_0/2v_t$. Since $s_0^3 = 2\sqrt{2}r^3$,

$$\boxed{G = \frac{3\,Q(r)}{r^3}, \qquad r = \frac{v_0}{2v_t} = \frac{eE_0}{2m\omega v_t},} \tag{11.12}$$

which is Eqs. (3.10) and (3.11) [9]. Silin writes

$$\nu = \frac{32\sqrt{2\pi}N_i e_i^2 e^2}{m^2 v_T^3}\Big(\frac{m\omega v_T}{eE_0}\Big)^3 Q(r)\ln\frac{k_{\max}}{k_{\min}} \tag{11.13}$$

with $e_i = Ze$ and $N_i = n_i$. Since $m\omega v_T/eE_0 = v_t/v_0 = 1/2r$ and $32\sqrt{2\pi}/8r^3 = \frac{4}{3}\sqrt{2\pi}\cdot 3/r^3$, his ν is $\nu_T\ln(k_{\max}/k_{\min})\cdot 3Q(r)/r^3$.

(5) The limits. For small s_0, with $\langle\cos^{2n}\theta\rangle = (2n-1)!!/(2n)!!$, the series of g gives $\langle g\rangle = \frac{2}{3}s_0^2 - \frac{3}{10}s_0^4 + \frac{5}{56}s_0^6$. For large s_0 write $\langle g(s_0\cos\theta)\rangle = (2/\pi)\int_0^{s_0} g(s)(s_0^2 - s^2)^{-1/2}\mathrm{d}s$ and split g into $\sqrt{\pi}\,\mathrm{erf}(s)/s$ and $-2e^{-s^2}$. With $\int_0^S \mathrm{erf}(s)/s\,\mathrm{d}s = \ln S + \gamma/2 + \ln 2 + o(1)$, $\int_0^1[(1-x^2)^{-1/2} - 1]\mathrm{d}x/x = \ln 2$ and $\int_0^\infty 2e^{-s^2}\mathrm{d}s = \sqrt{\pi}$, proved in Appendix B,

$$\langle g(s_0\cos\theta)\rangle = \frac{2}{\sqrt{\pi}\,s_0}\Big[\ln(4s_0) + \frac{\gamma}{2} - 1\Big] + o(s_0^{-1}). \tag{11.14}$$

Hence

$$\boxed{\begin{aligned} &G = 1 - \frac{9}{20}s_0^2 + \frac{15}{112}s_0^4 + O(s_0^6) \quad (s_0 \ll 1), \qquad G \simeq \frac{3}{\sqrt{\pi}s_0^3}\Big[\ln(4s_0) + \frac{\gamma}{2} - 1\Big] \quad (s_0 \gg 1), \\ &Q(r) \simeq \frac{\ln r + c_S}{2\sqrt{2\pi}}, \qquad c_S = \ln(4\sqrt{2}) + \frac{\gamma}{2} - 1 = 1.0215. \end{aligned}} \tag{11.15}$$

Silin found $Q(r) \simeq (\ln r + 1)/2\sqrt{2\pi}$, the 1 being, in his words, the result of numerical integration [9, Eq. (3.13)]. The constant is $c_S = \ln(4\sqrt{2}) + \gamma/2 - 1$.

11.4 The quantum strong-field result

(1) Table 4 compares the exact ratio $R(\beta, u)$ of Eq. (11.3) with Silin's $G(\beta/\sqrt{2})$ at $u = 0.3$. The two agree for weak fields and part for strong ones, where the quantum result is larger by a factor that grows slowly with β.

(2) The reason is the upper cutoff. In the quantum theory the momentum transfer is limited by backscattering of an electron from the moving ion, $\hbar k \lesssim 2m|\hat{\mathbf{k}}\cdot\mathbf{v}_{\rm osc}(t)|$, which grows with the field, while the lower cutoff stays at $k \sim \omega/v_t$ by step (3) of the quasi-static limit. At fixed logarithm both cutoffs are held. Figure 4 shows the ratio for three photon energies.

Table 4: The ratio $R = \nu_{\rm eff}(v_0)/\nu_{\rm eff}(0)$ of the exact quantum heating rate, Eq. (11.3) at $u = 0.3$, and Silin's ratio at fixed logarithm, Eq. (11.12), against $\beta = v_0/v_t$.

β	0.05	0.25	1	2	4	8
$R(\beta, 0.3)$	0.9996	0.9909	0.8673	0.5909	0.2102	0.0440
$G(\beta/\sqrt{2})$	0.9994	0.9861	0.8050	0.4604	0.1242	0.0224

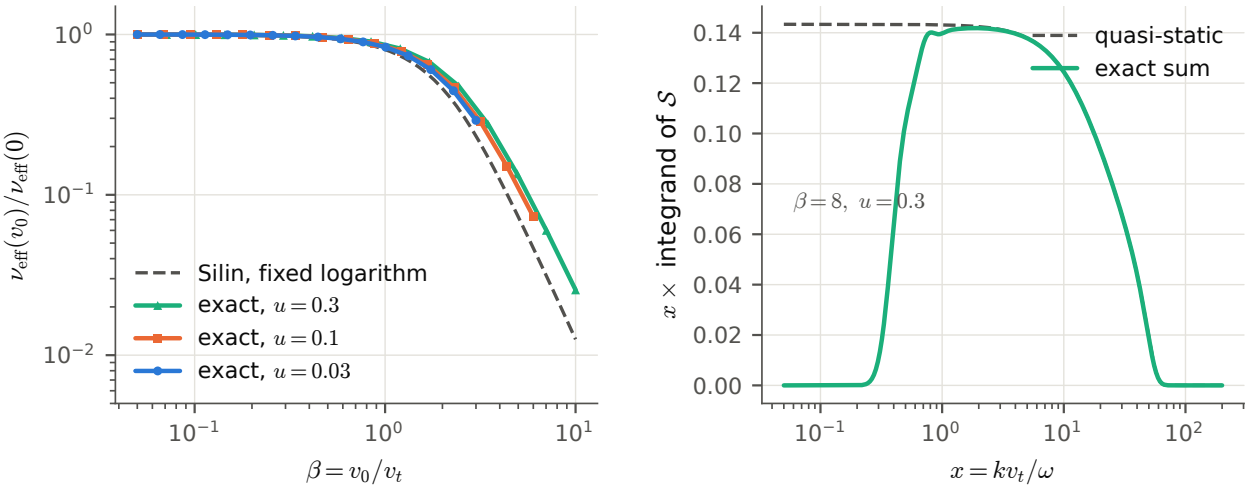


Figure 4: Strong fields. Left, the ratio $R = \nu_{\rm eff}(v_0)/\nu_{\rm eff}(0)$ of the exact quantum heating rate of Eq. (11.3) for three photon energies, and Silin's ratio G at fixed logarithm, Eq. (11.12). Right, the integrand of $\mathcal{S}$ times $x = kv_t/\omega$ at $\beta = 8$, $u = 0.3$, exact and in the quasi-static limit of Eq. (11.7). The exact integrand is cut off near $x \approx 0.3$ by the thermal factor, and both are cut off near $x \approx 2\beta/u$ by the quantum limit on the momentum transfer.

12 The kinetic equation of an isotropic distribution

This section returns to the master equation and follows the distribution of the electrons, not only its energy. Let $F(v)$ be the isotropic velocity distribution, $n_e = \int 4\pi v^2 F\,\mathrm{d}v$, which is $2m^3 f/(2\pi\hbar)^3$ in terms of the occupation numbers.

12.1 The averaged master equation for weak fields

(1) For weak fields the master equation (6.11) has the terms $\ell = 0, \pm 1$. The term $\ell = 0$ connects states of equal energy and vanishes for an isotropic distribution. The field generates an anisotropy proportional to $(\hat{\mathbf{p}} \cdot \hat{\mathbf{e}})^2 - \frac{1}{3}$, which elastic collisions remove faster than the field heats, and the isotropic part obeys the master equation averaged over the direction of $\mathbf{p}$,

$$\frac{\partial F(E)}{\partial t} = R_+(E)\big[F(E+\hbar\omega) - F(E)\big] + R_-(E)\big[F(E-\hbar\omega) - F(E)\big], \tag{12.1}$$

where $R_\pm = n_i v \bar{\sigma}_{\pm 1}$ with the cross sections of Eq. (5.20),

$$\boxed{R_\pm(E) = \frac{2\pi}{3}\,\frac{n_i Z^2 e^4 a_0^2 m}{\hbar^2 p}\,L_\pm, \qquad L_\pm = \ln\frac{p + p'_\pm}{|p - p'_\pm|}, \qquad R_-(E) = 0 \ \text{ for } \ E < \hbar\omega.} \tag{12.2}$$

(2) The density of states per unit energy and volume is $g(E) = 2 \cdot 4\pi p^2 (2\pi\hbar)^{-3} \mathrm{d}p/\mathrm{d}E = 8\pi m p/(2\pi\hbar)^3$, so $g(E)R_+(E)$ is proportional to $L_+(p)$, which is symmetric in p and p'_+. Since L_- at the momentum p'_+ is the same logarithm,

$$\boxed{g(E)\, R_+(E) = g(E+\hbar\omega)\, R_-(E+\hbar\omega).} \tag{12.3}$$

The upward rate from E and the downward rate from $E+\hbar\omega$ are balanced state by state, which is detailed balance for the rates.

12.2 The flux form and the Fokker–Planck limit

(1) Define the net flux of electrons per unit volume across the step from E to $E+\hbar\omega$,

$$\Phi(E) = g(E) R_+(E) \big[F(E) - F(E+\hbar\omega)\big]. \tag{12.4}$$

By Eq. (12.3) the averaged master equation is

$$g(E)\, \frac{\partial F(E)}{\partial t} = \Phi(E-\hbar\omega) - \Phi(E). \tag{12.5}$$

(2) For $\hbar\omega \ll E$ expand, $\Phi(E) \simeq -g R_+ \hbar\omega\, \partial_E F$ and $\Phi(E-\hbar\omega) - \Phi(E) \simeq -\hbar\omega\, \partial_E \Phi$, so $g\, \partial_t F \simeq \partial_E [g R_+ (\hbar\omega)^2 \partial_E F]$. The product $(\hbar\omega)^2 R_+ = \frac{2\pi}{3} n_i Z^2 e^4 v_0^2 L/v$ stays finite as $\hbar\omega \to 0$, because $(\hbar\omega)^2 a_0^2 = \hbar^2 v_0^2$.
(3) Change to the velocity, with $g \propto v$ and $\partial_E = (mv)^{-1}\partial_v$,

$$\frac{\partial F}{\partial t} = \frac{2\pi n_i Z^2 e^4 v_0^2}{3m^2}\, \frac{1}{v^2}\, \frac{\partial}{\partial v} \Big[\frac{L(v)}{v}\, \frac{\partial F}{\partial v} \Big]. \tag{12.6}$$

In terms of the momentum-transfer collision frequency $\nu_m(v) = 4\pi n_i Z^2 e^4 L(v)/m^2 v^3$,

$$\boxed{\frac{\partial F}{\partial t} = \frac{1}{v^2}\, \frac{\partial}{\partial v} \Big[v^2\, \frac{v_0^2}{6}\, \nu_m(v)\, \frac{\partial F}{\partial v} \Big], \qquad \nu_m(v) = \frac{4\pi n_i Z^2 e^4}{m^2 v^3}\, L(v), \qquad L(v) = \ln \frac{2mv^2}{\hbar\omega}.} \tag{12.7}$$

This is the inverse bremsstrahlung operator used by Langdon [20]. The quantum theory supplies its logarithm, $L = \ln(4E/\hbar\omega)$, which is the limit $\hbar\omega \ll E$ of $\ln[(p+p'_+)/(p'_+ - p)]$, the ratio of the largest momentum transfer $2p$ to the smallest $\hbar\omega/v$.

12.3 The heating rate of an arbitrary isotropic distribution

(1) Multiply Eq. (12.7) by $4\pi v^2\, mv^2/2$, integrate, and integrate by parts once. With $C = 2\pi n_i Z^2 e^4 v_0^2/3m^2$,

$$P = \int_0^\infty 4\pi v^2 \frac{mv^2}{2} \frac{\partial F}{\partial t} \mathrm{d}v = -4\pi m C \int_0^\infty L(v)\, \frac{\partial F}{\partial v}\, \mathrm{d}v. \tag{12.8}$$

(2) For a fixed logarithm the integral is $-LF(0)$, and

$$\boxed{P = 4\pi m C\, L\, F(0).} \tag{12.9}$$

The absorption is proportional to the number of slow electrons, measured by $F(0)$. For the Maxwellian $F(0) = n_e (2\pi v_t^2)^{-3/2}$, and $P/(n_e m v_0^2/2) = \nu_T L$, which is Eq. (8.4) with $\Lambda \to L$.

(3) With the velocity-dependent logarithm and the Maxwellian, $-\partial_v F = (v/v_t^2)F$, and $y = v^2/2v_t^2$ is exponentially distributed under the weight $(v/v_t^2)e^{-v^2/2v_t^2}$. The average of $L = \ln(4y/u)$ is $\ln(4/u) - \gamma$, which is Eq. (8.5). The Fokker–Planck limit reproduces the exact Maxwellian result to first order in u.
(4) The operator heats the slow electrons fastest, since $v^2\nu_m \propto v^{-1}$. When this heating outpaces the electron–electron collisions that restore the Maxwellian, the distribution becomes flat-topped [20]. For $F \propto \exp[-(v/v_m)^M]$ at fixed density and kinetic energy, $v_m = v_t[3\Gamma(3/M)/\Gamma(5/M)]^{1/2}$ and $F(0) = Mn_e/4\pi v_m^3\Gamma(3/M)$, so Eq. (12.9) at fixed L gives the ratio to the Maxwellian

$$\boxed{\frac{P_M}{P_2} = \frac{(2\pi)^{3/2}M}{4\pi\Gamma(3/M)}\left[\frac{\Gamma(5/M)}{3\Gamma(3/M)}\right]^{3/2} = 1,\ 0.621,\ 0.501,\ 0.446 \quad \text{for} \quad M = 2, 3, 4, 5,} \tag{12.10}$$

and 0.336 for $M \to \infty$. This is consistent with the reduction by up to a factor of two that Langdon reports [20].

13 Summary

In Summary: An electron in the laser field is a Volkov state, Eq. (4.5). Its matrix element with the ion potential is a sum of harmonics weighted by $J_\ell(\mathbf{k}\cdot\mathbf{a}_0)$, Eq. (5.4), and the golden rule gives the multiphoton cross section of Bunkin and Fedorov, Eq. (5.13). The operator method of the quantum Boltzmann equation paper turns these rates into the master equation (6.11) for the occupation numbers, exactly at second order in the ion potential and without Pauli blocking. Its energy moment is the heating rate, which in terms of the Lindhard function is Eq. (7.4), and in the oscillating frame of the interacting electron gas it becomes the dielectric form Eq. (10.9), with Dawson and Oberman for weak fields and Silin for strong ones.

For weak fields and Maxwell electrons the collision frequency of the Drude model is $\nu_T\Lambda(u)$ with $\Lambda(u) = [(1-e^{-u})/u]e^{u/2}K_0(u/2)$, which is $\ln(4k_BT/e^\gamma\hbar\omega)$ for $\hbar\omega \ll k_BT$. The Coulomb logarithm needs no cutoff. It is $\pi/\sqrt{3}$ times the thermally averaged Born Gaunt factor, Eq. (9.11), Kirchhoff's law holds exactly, and the frequency-integrated Gaunt factor is $2\sqrt{3}/\pi$. The degenerate gas gives Ziman's formula, and every distribution has the same tail for $\hbar\omega \gg k_BT$. Dynamic screening lowers the logarithm by 0.10 at a quarter of the critical density and by at most 0.56 as $\omega \to \omega_p$, Table 3. In strong fields the classical limit is Silin's $3Q(r)/r^3$, whose constant is $\ln(4\sqrt{2}) + \gamma/2 - 1$, and the quantum result exceeds it because the largest momentum transfer grows with the quiver velocity. For an isotropic distribution the master equation reduces to Langdon's operator with the quantum logarithm $\ln(2mv^2/\hbar\omega)$, Eq. (12.7).

Three corrections to the classical treatment of [1] follow, Appendix A. The free–free and the Drude absorption coefficients, Eqs. (11.31) and (11.34) [1], differ by a factor of 4.4 as printed, and they agree only when the Gaunt factor is $(\sqrt{3}/\pi)\ln\Lambda$. The current of Eq. (11.8) [1] carries the wrong sign, which Eq. (11.9) [1] does not inherit. The constant of the Coulomb logarithm of Eq. (10.136) [1] is 0.9 below the value of the cutoffs stated with it.

The theory has three limits that the next steps of this project should lift. The Born approximation needs $Ze^2/\hbar v \ll 1$, which fails for high Z and low temperature, where the exact Coulomb Gaunt factors of Sommerfeld [21] and Karzas and Latter [22] and the Elwert factor [23] replace the Born ones [24]. The random-phase response omits local-field corrections and the correlations of a strongly coupled plasma. The anisotropy of the distribution and its relaxation by electron–electron collisions, which decide when Langdon's flat-topped distribution forms, require the full kinetic equation with both collision terms.

A Reduction to Atzeni and Meyer-ter-Vehn

This appendix takes the equations of [1] that bear on inverse bremsstrahlung in turn, and states for each what the quantum theory gives. Table 5 collects the numbers, which the check script recomputes from the constants of CODATA 2018.

A.1 The Drude model, Eqs. (11.7) to (11.12)

Section 2.1 derives Eqs. (11.7), (11.9) and (11.12) [1]. The current of Eq. (11.8) [1] as printed is $e^2 n_0 \mathbf{E}_1/[i m_e(\omega + i\nu_e)]$, which equals $-\sigma\mathbf{E}_1$ with the conductivity σ of Eq. (2.3). Inserted into Eq. (11.4) [1] it would give $\varepsilon_T = 1 + \omega_p^2/\omega(\omega + i\nu)$, the wrong sign of the plasma term. Eq. (11.9) [1] has the correct sign, so the slip is confined to Eq. (11.8). The quantum theory replaces ν_e by ν_{eff} of Eq. (7.11).

A.2 The collision time and the Coulomb logarithm, Eqs. (10.133) and (10.136)

(1) Eq. (10.133) [1] is $\tau_e = 1/\nu_T \ln\Lambda_e$ with ν_T of Eq. (2.10). Its numerical coefficient, 1.088×10^{-11} s, agrees with the printed 1.09×10^{-11} s.
(2) Eq. (10.136) [1] prints $\ln\Lambda_e = 7.1 - 0.5\ln n_e + \ln T_e$. The text states that it is the logarithm of $\lambda_D/\lambda_{\mathrm{dB}}$ with $\lambda_{\mathrm{dB}} = \hbar/2m_e v$ and $m_e v^2/2 = 3k_B T_e/2$. At $k_B T_e = 1$ keV and $n_e = 10^{21}$ cm^{-3} these cutoffs give $\lambda_D = 7.43\times10^{-7}$ cm, $\lambda_{\mathrm{dB}} = 2.52\times10^{-10}$ cm and $\ln(\lambda_D/\lambda_{\mathrm{dB}}) = 7.99$, so the printed constant is 0.89 lower than its stated definition.
(3) The quantum theory gives the logarithm of Eq. (8.4), whose cutoffs are v_t/ω and $0.445\hbar/mv_t$, Eq. (8.6), and Eq. (10.13) with screening. Table 1 compares them with Eq. (10.136).

A.3 The critical density and Eqs. (11.30) to (11.33)

(1) Eq. (11.30) [1] is $n_c = \pi m_e c^2/e^2\lambda_L^2 = 1.115\times10^{21}/\lambda_L^2$ cm^{-3} with λ_L in μm, printed as 1.1×10^{21}.
(2) Eq. (11.33) [1] prints $\nu_{ei}/c \approx 25 Z_i/\lambda_L^2 T_e^{3/2}$ cm^{-1} for $\ln\Lambda_e \approx 8$. Equations (10.133) and (11.30) [1] give 27.3 for the same logarithm.
(3) The text estimates $\ln\Lambda_e \approx 8$ from Eq. (10.136) for $k_B T_e = 1$ keV at the critical density of $\lambda_L = 0.5$ μm, $n_c = 4.46\times10^{21}$ cm^{-3}. Eq. (10.136) gives 6.35 there and its stated cutoffs 7.24. The quantum theory gives $\Lambda(u) = 6.81$ at $u = \hbar\omega/k_B T = 0.00248$ without screening.

A.4 Kramers' cross section and the free–free opacity, Eqs. (10.89) to (10.93)

(1) The Maxwell average of Kramers' cross section, Eq. (10.89) [1], multiplied by $h\nu/4\pi A m_p$, is

$$\begin{aligned}\eta_\nu &= \frac{h\nu}{4\pi A m_p}\, n_e \Big(\frac{m}{2\pi k_B T}\Big)^{3/2} 4\pi \,\frac{32\pi^2 Z^2 e^6}{3\sqrt{3} m^2 c^3 h\nu} \int_{v_{\min}}^{\infty} v\, e^{-mv^2/2k_B T}\,\mathrm{d}v \\ &= \frac{16\pi}{3\sqrt{6\pi}}\,\frac{e^6 Z^2 n_e}{m^2 c^3 (k_B T/m)^{1/2} A m_p}\, e^{-h\nu/k_B T},\end{aligned} \tag{A.1}$$

which is Eq. (10.91) [1]. Kirchhoff's law with $B_\nu = 2h\nu^3 c^{-2} e^{-h\nu/k_B T}$ for the true absorption gives Eq. (10.92) [1] in its symbolic form. Its numerical form evaluates to $2.74\, Z^3\rho/A^2 T^{1/2}(h\nu)^3$ cm^2 g^{-1} against the printed 2.78, and Eq. (10.93) [1] to 1.74×10^{17} against the printed 1.76×10^{17} W g^{-1}. The two differences are the same 1.4 percent.
(2) The Gaunt factor $g \approx 2\sqrt{3}/\pi$ that the text applies to Eq. (10.93) [1] is the power-averaged Born Gaunt factor of Eq. (9.10), exactly.

A.5 The two absorption coefficients, Eqs. (11.31) and (11.34)

(1) By Eq. (9.14) the net absorption coefficient of Kramers' form with a Gaunt factor g is $K = (\nu_T/c)(\omega_p^2/\omega^2)(\pi/\sqrt{3})\, g\,[(1-e^{-u})/u]/n_r$. With $g = 1$ this is Eq. (11.34) [1], exactly. The Drude form Eq. (11.31) [1] is the same with $(\pi/\sqrt{3})g(1-e^{-u})/u$ replaced by $\ln\Lambda_e$. Their ratio is

$$\frac{K_{(11.31)}}{K_{(11.34)}} = \frac{\ln\Lambda_e}{(\pi/\sqrt{3})\, g\,(1-e^{-u})/u}. \tag{A.2}$$

(2) At $\lambda_L = 0.5$ μm and $k_BT = 1$ keV, $u = 0.00248$, and with $g = 1$ and $\ln\Lambda_e = 8$ the ratio is 4.42. The numerical forms give 6.19 and 27.3 $Z/\lambda_L^2 T_e^{3/2}$ cm^{-1} at $n_e = n_c$, the same ratio. The statement below Eq. (11.34) [1], that the two agree within a few percent, holds only for $g = (\sqrt{3}/\pi)\ln\Lambda_e = 4.4$. (3) The quantum theory gives both at once. The Born Gaunt factor at $u = 0.00248$ is $\langle g\rangle = 3.758$, and both forms become $\nu_T\Lambda(u)$ with $\Lambda = 6.81$.

A.6 The classical dielectric function, Eq. (11.19)

The classical limit of the Lindhard function of Maxwell electrons is Eq. (11.19) [1], Section 10.5 step (2). Its imaginary part is Eq. (7.8).

A.7 Bohr and Bethe, Eqs. (11.72) to (11.74)

For stopping, [1] takes the smaller of the classical logarithm of Bohr and the quantum one of Bethe, Eq. (11.74). The same choice arises here. The quantum Coulomb logarithm of Eq. (8.4) is the Bethe-type result and holds for $\eta = Ze^2/\hbar v_t \ll 1$. For $\eta \gtrsim 1$ the lower cutoff becomes the classical distance $r_0 = Ze^2/mv^2$ and the Born result overestimates the absorption, which the exact Coulomb Gaunt factors correct.

B Mathematical identities

B.1 An integral of the error function

Integrate by parts, $\int_0^S \mathrm{erf}(s)s^{-1}\mathrm{d}s = \mathrm{erf}(S)\ln S - (2/\sqrt{\pi})\int_0^S e^{-s^2}\ln s\,\mathrm{d}s$. Differentiating $\int_0^\infty s^{2a-1}e^{-s^2}\mathrm{d}s = \Gamma(a)/2$ with respect to a at $a = \frac12$ gives $\int_0^\infty e^{-s^2}\ln s\,\mathrm{d}s = \Gamma'(\frac12)/4 = \frac{\sqrt{\pi}}{4}\psi(\frac12) = -\frac{\sqrt{\pi}}{4}(\gamma + 2\ln 2)$, with $\psi(\frac12) = -\gamma - 2\ln 2$ from Eq. (5.4.13) [11]. Since $1 - \mathrm{erf}\,S$ and the tail of the last integral decay faster than any power,

$$\int_0^S \frac{\mathrm{erf}\,s}{s}\,\mathrm{d}s = \ln S + \frac{\gamma}{2} + \ln 2 + o(1). \tag{B.1}$$

B.2 A logarithmic integral

With $x = \sin\varphi$,

$$\int_0^1 \Big[\frac{1}{(1-x^2)^{1/2}} - 1\Big]\frac{\mathrm{d}x}{x} = \int_0^{\pi/2}\frac{1-\cos\varphi}{\sin\varphi}\,\mathrm{d}\varphi = \int_0^{\pi/2}\tan\frac{\varphi}{2}\,\mathrm{d}\varphi = \Big[-2\ln\cos\frac{\varphi}{2}\Big]_0^{\pi/2} = \ln 2. \tag{B.2}$$

Table 5: Equations of [1] on inverse bremsstrahlung, as printed and as recomputed or derived here.

Equation	Printed	Recomputed or derived	Status
(11.8)	$e^2 n_0 \mathbf{E}_1/[im_e(\omega + i\nu_e)]$	$ie^2 n_0 \mathbf{E}_1/[m_e(\omega + i\nu_e)]$	sign slip
(11.9), (11.12)	Drude ε_T and K	Eqs. (2.4), (2.9)	confirmed
(10.133)	1.09×10^{-11} s	1.088×10^{-11} s	confirmed
(10.136)	constant 7.1	7.99 from its cutoffs	constant low by 0.89
(11.30)	$1.1 \times 10^{21}/\lambda_L^2$	$1.115 \times 10^{21}/\lambda_L^2$	confirmed
(11.33)	25	27.3	9 percent low
text, $\ln \Lambda_e \approx 8$	from (10.136)	6.35 from (10.136), 6.81 quantum	inconsistent
(10.91)	symbolic form	Maxwell average of (10.89)	confirmed
(10.92)	2.78	2.74	1.4 percent
(10.93)	1.76×10^{17}	1.74×10^{17}	1.4 percent
$g \approx 2\sqrt{3}/\pi$	quoted for (10.93)	Born power average, Eq. (9.10)	exact
(11.31) vs (11.34)	agree within a few percent	ratio 4.42	needs $g = 4.4$
(11.19)	classical ε_L	classical limit of Lindhard	confirmed

B.3 The large-field limit of the phase average

(1) With $s = s_0 \sin\theta'$ and the symmetry of the average, $\langle g(s_0 \cos\theta)\rangle = (2/\pi)\int_0^{s_0} g(s)(s_0^2 - s^2)^{-1/2}\mathrm{d}s$.
(2) For the part $\sqrt{\pi}\,\mathrm{erf}(s)/s$ write $(s_0^2 - s^2)^{-1/2} = s_0^{-1}\{1 + [(1 - s^2/s_0^2)^{-1/2} - 1]\}$. The first term gives $s_0^{-1}\sqrt{\pi}[\ln s_0 + \gamma/2 + \ln 2]$ by Eq. (B.1). In the second term $\mathrm{erf}(s) \to 1$ wherever the bracket is not small, and Eq. (B.2) gives $s_0^{-1}\sqrt{\pi}\ln 2$.
(3) The part $-2e^{-s^2}$ is confined to $s \lesssim 1$, where $(s_0^2 - s^2)^{-1/2} \simeq s_0^{-1}$, and gives $-s_0^{-1}\sqrt{\pi}$.
(4) Adding, $\langle g\rangle = (2/\pi s_0)\sqrt{\pi}[\ln s_0 + \gamma/2 + 2\ln 2 - 1] = (2/\sqrt{\pi}s_0)[\ln(4s_0) + \gamma/2 - 1]$, which is the form used in Eq. (11.15).

B.4 The phase average of a Gaussian

With $\cos^2\theta = \frac{1}{2}(1 + \cos 2\theta)$ and Eq. (10.32.1) [11], $I_0(z) = \pi^{-1}\int_0^\pi e^{\pm z\cos\phi}\mathrm{d}\phi$,

$$\left\langle e^{-a\cos^2\theta}\right\rangle_\theta = e^{-a/2}\left\langle e^{-(a/2)\cos 2\theta}\right\rangle_\theta = e^{-a/2} I_0\Big(\frac{a}{2}\Big). \tag{B.3}$$

C The numerical checks

The script `detailed_derivation_check.py` verifies every boxed result of these notes, in their order, by an independent numerical route. The routes are quadrature of the defining integrals in place of the closed forms, finite differences in place of derivatives, the golden-rule time integral at finite times, the cross-section route against the Lindhard route, an exact computation in the Fock space of three fermion modes for the equation of motion, and the published equations of [3, 9, 1] as printed. It runs with `numpy` and `scipy` under `/usr/bin/python3` in about fifteen seconds. Table 6 is written by the script itself. A quantitative check passes when its deviation is below the tolerance

stated in the script, which reflects the precision of the quadrature or of the asymptotic expansion it tests. A printed number passes when the script reproduces it to its last printed digit. The figures are made by `detailed_derivation_fig.py`.

Table 6: The numerical checks of `detailed_derivation_check.py`, by the equation, table or section they verify. The second column counts all checks of the row. The third gives the largest deviation among its quantitative checks, relative, or absolute for quantities near zero, and is empty when the row has none. The fourth counts the numbers printed in these notes that the script reproduces to their last printed digit, and the fifth the logical conditions, such as a sign or a monotone decrease.

Equation	Checks	Largest deviation	Printed	Logical
Eq. (2.3)	2	1.3×10^{-16}		
Eq. (2.4)	2	2.9×10^{-16}		
Eq. (2.5)	1	6.4×10^{-16}		
Eq. (2.7)	2	2.9×10^{-11}		
Eq. (2.9)	3	1.6×10^{-7}		
Eq. (2.10)	1		1	
Eq. (3.3)	3	1.6×10^{-12}		
Eq. (3.4)	2	0		
Eq. (3.11)	2	1.9×10^{-16}		
Eq. (3.7)	3	1.8×10^{-16}		
Eq. (4.5)	2	5.0×10^{-13}		
Eq. (4.7)	3	4.6×10^{-16}		
Eq. (4.9)	9	5.6×10^{-16}		
Eq. (4.11)	6	8.5×10^{-15}		
Eq. (4.10)	2	2.5×10^{-7}		
Eq. (5.4)	2	1.2×10^{-16}		
Eq. (5.9)	2	1.1×10^{-3}		
Eq. (5.11)	1	1.4×10^{-3}		
Eq. (5.13)	2	8.6×10^{-8}		
Eq. (5.14)	2	4.0×10^{-16}		
Eq. (5.19)	3	6.1×10^{-15}		
Eq. (5.20)	3	2.0×10^{-15}		
Eq. (5.21)	15	1.5×10^{-6}		1
Eq. (5.23)	2	7.9×10^{-6}		1
Eq. (6.4)	2	0		1
Eq. (6.6)	1	1.8×10^{-16}		
Eq. (6.10)	2	1.3×10^{-11}		
Eq. (6.8)	1	3.0×10^{-13}		
Eq. (6.12)	2	3.6×10^{-16}		
Eq. (6.11)	1	1.4×10^{-14}		
Eq. (6.14)	2	1.5×10^{-15}		1
Eq. (7.5)	3	3.7×10^{-10}		
Eq. (7.7)	3	3.3×10^{-15}		
Eq. (7.6)	6	2.3×10^{-14}		
Eq. (7.8)	4	7.0×10^{-8}		2

Equation	Checks	Largest deviation	Printed	Logical
Eq. (8.3)	2	1.7×10^{-15}		
Eq. (8.4)	10	6.8×10^{-12}		
Eq. (8.5)	4	2.5×10^{-3}		2
Eq. (8.6)	4		4	
Table 2	36		36	
Eq. (8.7)	5	4.5×10^{-6}		
Eq. (8.11)	8	2.4×10^{-7}		
Eq. (8.12)	6	5.0×10^{-4}		2
Eq. (8.15)	5	8.4×10^{-5}		
Eq. (9.2)	1	1.1×10^{-16}		
Eq. (9.4)	1	1.7×10^{-15}		
Eq. (9.5)	3	2.4×10^{-12}		
Eq. (9.6)	3	2.2×10^{-16}		
Eq. (9.9)	4	3.5×10^{-12}		
Eq. (9.11)	4	5.2×10^{-15}		
Eq. (9.10)	3	1.1×10^{-13}	1	
Eq. (9.14)	2	6.9×10^{-16}		
Eq. (10.4)	1	6.6×10^{-16}		
Eq. (10.11)	5	1.5×10^{-11}		
Eq. (10.9)	4	1.1×10^{-15}		
Eq. (10.13)	8	1.3×10^{-3}		
Eq. (10.10)	1	3.1×10^{-15}		
Table 3	26		26	
Eq. (11.5)	12	9.6×10^{-16}		
Eq. (11.3)	4	1.5×10^{-5}		
Table 4	12		12	
Eq. (11.7)	2	3.9×10^{-5}		1
Eq. (11.8)	7	8.7×10^{-10}		
Eq. (11.12)	8	2.4×10^{-14}		
Eq. (11.9)	1	4.5×10^{-7}		
Eq. (11.15)	6	1.0×10^{-4}	1	
Eq. (12.3)	3	0		
Eq. (12.2)	3	0		
Eq. (12.7)	4	2.8×10^{-8}		2
Eq. (12.9)	2	2.2×10^{-13}		
Eq. (12.10)	7	3.0×10^{-16}	5	
Table 5	15	3.5×10^{-8}	13	
Sec. A.2	4		4	
Sec. A.5	7	3.3×10^{-16}	5	
Table 1	36		36	
Sec. 3.4	2		2	
Sec. 8	2		2	
Eq. (B.1)	2	1.8×10^{-16}		
Eq. (B.2)	1	4.5×10^{-14}		
Sec. B	6	7.1×10^{-6}		

Equation	Checks	Largest deviation	Printed	Logical
Total	394		148	13

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